SSC JE 2026

COMPLETE BASIC ELECTRICAL & CIRCUIT THEORY

Detailed theory | Formulas and derivations | Step-by-step solved models | Practice questions

EXAM COVERAGE	IN THIS BOOK
Paper I and Paper II	Conceptual and numerical preparation
Direct 2026 syllabus	Fundamentals, network solution methods and theorems, magnetic circuits, AC circuits, resonance, balanced 3-phase circuits and complex power
Useful extensions	RC/RL transients, source transformation, bridging methods, practical battery models, power-factor correction
Study format	Theory → derivation/shortcut → worked problems → question-pattern traps → self-test

Prepared for focused SSC Junior Engineer (Electrical) revision. This is an original study guide, not an official SSC publication.
Primary reference: Staff Selection Commission, Junior Engineer Examination 2026, 02 September 2026, paragraph 14.3.2, printed page 26 of 97.
Official syllabus:
https://ssc.gov.in/api/attachment/uploads/masterData/NoticeBoards/Notice_of_adv_je_2026.pdf
Version: 11 October 2026 • Equations use SI units unless noted.

How to use these notes

Read the principles once, then write out every worked solution without looking. After completing a chapter, attempt its linked practice questions before moving on. Use the formula sheets for speed; use the longer derivations when you cannot explain why a formula works.

SCOPE CHECK 2026 direct syllabus: resistor/capacitor/inductor/battery fundamentals; voltage/current/power/energy/flux; KCL, KVL, node, mesh; Thevenin, Norton, superposition, maximum power transfer; flux, reluctance, MMF, self and mutual inductance; AC RMS/average, series and parallel AC circuits, resonance, balanced three-phase, star-delta transformation, complex power, power factor.

SYLLABUS BOUNDARY RC/RL switching transients and some special circuits are useful extensions and previous-style practice; they are not separately named under the 2026 Paper II electrical core.

Notation: V = voltage in volt; I = current in ampere; R = resistance in ohm (Ω); C = capacitance in farad; L = inductance in henry; f = frequency in hertz; ω = angular frequency in rad/s; Φ = flux in weber; B = flux density in tesla; $j^2 = -1$.

Phasors in these notes use **RMS magnitudes** unless a question explicitly states peak values. For a passive load, positive reactive power Q means inductive (lagging) and negative Q means capacitive (leading).

Contents

01 Electrical quantities, sources and power 5

1.1 Charge, current, voltage and electric power . . . 5
1.2 Ideal and practical circuit elements . . . 6
1.3 Ohm law, active/passive and linear/nonlinear . . . 6
1.4 Efficiency and load matching . . . 6

02 Resistance, DC networks and bridge circuits 7

2.1 Material resistance, temperature, wire stretching . . . 7
2.2 Series and parallel resistors . . . 8
2.3 Star–delta transformations (resistive) . . . 8
2.4 Wheatstone bridge and symmetry . . . 9
2.5 Open/short identification and equivalent resistance . . . 9

03 Capacitors and electrostatic energy 10

3.1 Capacitance and its controlling factors . . . 10
3.2 Inserting a dielectric: disconnected versus connected . . . 10
3.3 Capacitor DC behavior and breakdown . . . 11

04 Inductors, coupling, batteries and source models 12

4.1 Inductance and magnetic energy . . . 12
4.2 Mutual inductance and dot convention . . . 12
4.3 Batteries, EMF, terminal voltage . . . 12
4.4 Source transformations . . . 13

05 KCL, KVL, nodal and mesh analysis 14

5.1 Kirchhoff laws and reference directions . . . 14
5.2 Nodal analysis: general recipe . . . 14
5.3 Supernode method . . . 14
5.4 Mesh-current analysis . . . 15
5.5 Supermesh and current sources . . . 15
5.6 Consistency checks: power balance and units . . . 15

06 Network theorems and maximum power 16

6.1 Thevenin theorem: exact five-step procedure . . . 16
6.2 Norton theorem and source conversion . . . 16
6.3 Superposition theorem . . . 17
6.4 Maximum power transfer . . . 17
6.5 Special source and theorem traps . . . 17

07 Magnetic circuits, materials and induction 19

7.1 Magnetic quantities and analogy with electricity . . . 19

7.2 Air gaps, leakage and fringing . . . 19
7.3 Field formulas: wire, solenoid, toroid . . . 19
7.4 Magnetic materials and B-H curve . . . 20
7.5 Faraday law, Lenz law and motional EMF . . . 20
7.6 Self-inductance from magnetic geometry . . . 20

08 AC waveforms, RMS, average values and phasors 21

8.1 Sinusoidal waveform and angular relationships . . . 21
8.2 RMS definition and derivation . . . 21
8.3 Average value and waveform factors . . . 21
8.4 Phasor basics and complex arithmetic . . . 22
8.5 Lead/lag quick reference . . . 22

09 AC circuits, impedance, power and correction 23

9.1 R, L and C impedance . . . 23
9.2 Series R-L, R-C, R-L-C circuits . . . 23
9.3 Parallel AC circuits and admittance . . . 24
9.4 Real, reactive, apparent and complex power . . . 24
9.5 Power-factor correction with shunt capacitor . . . 25
9.6 Exam-safe procedure for any AC circuit . . . 26

10 Resonance, bandwidth and tank circuits 27

10.1 Series RLC resonance . . . 27
10.2 Exact half-power frequencies . . . 27
10.3 Parallel resonance and ideal tank circuit . . . 28
10.4 High-frequency / low-frequency behavior and Q . . . 28

11 First-order RC and RL switching response (extension) 29

11.1 Universal first-order step response . . . 29
11.2 RC charging and discharging . . . 29
11.3 RL current growth and decay . . . 30
11.4 Immediately after switching: critical rules . . . 30

12 Balanced three-phase circuits and star–delta loads 31

12.1 Three-phase generation and balance . . . 31
12.2 Star (Y) connection . . . 31
12.3 Delta (Δ) connection . . . 31
12.4 Three-phase active, reactive and apparent power . . . 32
12.5 Star–delta impedance transformations . . . 32
12.6 Resistor heating and loads rated by line/phase voltage . . . 32
12.7 Neutral current and unbalanced cases . . . 33

13 Calculation models and exam decision tree 34

13.1 What to do first when you see a circuit . . . 34
13.2 Twelve frequent question models with one-line tricks . . . 34
13.3 Mixed worked models . . . 34
13.4 Careful negative-sign interpretation . . . 36

14 High-yield theory checkpoints 37

14.1 Which facts should be remembered exactly? . . . 37
14.2 Source and polarity reasoning . . . 37
14.3 Solve numericals without a calculator when possible . . . 37

15 All-in-one formula revision sheets 38

15.1 Basic quantities and DC networks . . . 38
15.2 Magnetism and induction . . . 38
15.3 Alternating current, impedance, resonance . . . 39
15.4 Three-phase systems . . . 39
15.5 Time-domain optional formulas . . . 39

16 Topic-wise practice set: objective models 40

17 Practice answer key with short explanations 45

18 Final checklist, mock strategy and common errors 49

18.1 Concept checklist: tick only when you can solve independently . . . 49
18.2 Most avoidable numerical mistakes . . . 50
18.3 Suggested 4-session revision cycle . . . 50
18.4 Sources and scope declaration . . . 50

01 Electrical quantities, sources and power

1.1 Charge, current, voltage and electric power

Electric charge (q) is measured in coulombs (C). Current is the time rate of flow of charge. Conventional current points in the direction of positive-charge flow, opposite to electron drift in a metal. **Potential difference (V)** is work done per unit charge. EMF refers to energy supplied per unit charge by a source; terminal voltage can differ from source EMF because of internal losses.

> **$I = dq/dt$ | For constant current: $q = It$ | $V = dW/dq$ | $P = VI = dW/dt$**

Use passive sign convention: an element **absorbs** power if current enters its marked positive-voltage terminal; $p=+vi$. If current enters the negative terminal, that element delivers power under the marked polarities.

> **$P = VI = I^2R = V^2/R$ (Ohmic resistor) | $W = \int p\,dt$ | 1 kWh = 3.6×10^6 J**

- For a device drawing 10 A from a 230 V DC supply: P=2300 W. For 2 h: E=4.6 kWh, not 4.6 kW.
- Rate of change matters: with $i(t)=3t^2$ A, q transferred from t=0 to t=2 s is $\int_0^2 3t^2\,dt=8$ C.

WORKED MODEL 01 | Nonconstant current and charge

Question: Current is $i(t)=4+2t$ A from $t=0$ to 3 s. Find transferred charge.

1. Integrate current over the interval: $q=\int_0^3(4+2t)dt$.
2. Evaluate $q=[4t+t^2]_0^3=12+9$.

> **RESULT Answer:** 21 C.

WORKED MODEL 02 | Electrical energy billing

Question: A 1.5 kW load operates for 4 h/day for 20 days. Determine units consumed and energy charge at Rs 7 per unit.

1. Total hours$=4\times20=80$ h.
2. Electrical energy$=1.5\times80=120$ kWh.
3. Charge$=120\times7=840$ rupees; fixed charges/taxes are excluded.

> **RESULT Answer:** 120 units (kWh); Rs 840.

1.2 Ideal and practical circuit elements

Element	Constitutive equation	Key fact
Resistor	$v=Ri$	Dissipates power; no energy storage in the ideal model
Capacitor	$i=C\,dv/dt$	Stores electric-field energy; v cannot jump at finite current
Inductor	$v=L\,di/dt$	Stores magnetic-field energy; i cannot jump at finite voltage
Ideal voltage source	v=prescribed	Internal resistance=0
Ideal current source	i=prescribed	Internal resistance=infinite

Ideal conductors are equipotential; an ideal open has zero current; an ideal short has zero voltage. **No** assertion about current in a short circuit or voltage across an open is possible without the surrounding network.

1.3 Ohm law, active/passive and linear/nonlinear

Ohm law $V=IR$ applies to an ohmic element at a specified physical condition (especially temperature). An element is linear if superposition applies to its voltage-current relation. A resistor whose resistance varies with temperature may have a nonlinear V-I curve even though $R=V/I$ can be computed at an operating point.

> **EXAM TRAP** An ideal short does not necessarily carry infinite current. It carries whatever current is imposed by the remaining circuit and source limitations.

1.4 Efficiency and load matching

> **η = useful output/input × 100% | Losses = input − output**

WORKED MODEL 03 | Simple efficiency model

Question: A conversion device has 2 kW electrical input and 360 W loss. Find efficiency.

1. Useful output$=2000-360=1640$ W.
2. Efficiency$=1640/2000=0.82$.

> **RESULT Answer:** 82%.

02 Resistance, DC networks and bridge circuits

2.1 Material resistance, temperature, wire stretching

$R = \rho\ell/A$ | $G=1/R$ | $\sigma=1/\rho$ | $R_2=R_1[1+\alpha_1(T_2-T_1)]$

Here ρ=resistivity ($\Omega\cdot$m), ℓ=length (m), A=cross-section (m^2), G=conductance (siemens), σ=conductivity (S/m), α_1=temperature coefficient referenced to T_1. For metals α is usually positive in a moderate temperature range; semiconductor behavior depends on material and operating point.

Wire problems often assume **constant material volume**, so $A\ell$ remains constant. If stretched to k times its initial length, area becomes A/k and resistance becomes k^2R. If a wire is divided into n equal pieces, each piece has resistance R/n; n identical pieces in parallel have equivalent R/n^2.

Constant-volume stretch: $R_{new}/R_{old} = (\ell_{new}/\ell_{old})^2$ | n equal wires in parallel: $R_{eq} = R/n^2$

WORKED MODEL 04 | Stretching wire

Question: A uniform wire has resistance 8 Ω and is stretched to 150% of its original length without volume change. Find the new resistance.

1. Length factor k=1.5.
2. Volume constant gives area factor=1/1.5.
3. $R_{new}=R_{old}\,k^2=8\times2.25$.

RESULT Answer: 18 Ω.

WORKED MODEL 05 | Wire cutting and reconnection

Question: A 36 Ω wire is cut into 3 equal lengths and the pieces connected in parallel.

1. Each piece has $R_{piece}=36/3=12$ Ω.
2. Three equal 12 Ω pieces in parallel: $R_{eq}=12/3$.

RESULT Answer: 4 Ω; direct shortcut $R/n^2=36/9$.

WORKED MODEL 06 | Temperature coefficient

Question: A conductor is 20 Ω at 20°C with $\alpha_{20}=0.004$/°C. Find resistance at 70°C.

1. Temperature rise=70−20=50°C.
2. $R_{70}=20[1+0.004\times50]=20\times1.2$.

RESULT Answer: 24 Ω.

2.2 Series and parallel resistors

Connection	General rule	Equal elements
Series	Req=R₁+R₂+...; same I	n identical resistors: nR
Parallel	1/Req=1/R₁+...; same V	n identical resistors: R/n
Two in parallel	Req=R₁R₂/(R₁+R₂)	R/2 if R₁=R₂
Voltage divider	Vout=Vin Rpart/Rseries	Apply only to a true series path
Current divider (2)	I₁=Itotal R₂/(R₁+R₂)	I/2 for equal branches

Loaded voltage divider: if a load RL is connected across R₂, first replace R₂ by R₂ || RL. Then Vout=Vin(R₂ || RL)/(R₁+R₂ || RL). The unloaded formula is not valid once an additional branch draws current.

WORKED MODEL 07 | Loaded voltage divider

Question: A 12 V source feeds 2 kΩ series resistor and a 4 kΩ resistor to ground. A 4 kΩ load is connected across the lower resistor.

1. Parallel bottom resistance: 4k || 4k=2 kΩ.
2. New divider: Vout=12×2/(2+2).
3. Load current=6V/4kΩ.

> **RESULT Answer:** Vout=6 V, Iload=1.5 mA. Without load Vout would be 8 V.

WORKED MODEL 08 | Current division

Question: A 12 A current source feeds parallel 3 Ω and 6 Ω resistors. Find branch currents.

1. Current in 3 Ω branch: I₃=12×6/(3+6).
2. Current in 6 Ω branch: I₆=12×3/(3+6).
3. KCL check: 8+4=12 A.

> **RESULT Answer:** 8 A through 3 Ω and 4 A through 6 Ω.

2.3 Star–delta transformations (resistive)

Label delta sides RAB, RBC and RCA. Label star arms RA, RB and RC connected from terminals A, B and C to the common centre. A delta can be converted to a star with each arm equal to the product of its adjacent delta sides divided by the sum of all delta sides.

RA = RAB·RCA / ΣRΔ | RB = RAB·RBC / ΣRΔ | RC = RBC·RCA / ΣRΔ

RAB = RA+RB+RA·RB/RC (cyclic permutations for RBC and RCA)

These relationships apply equally to impedances in sinusoidal steady state, replacing R by complex Z, provided the algebra is valid.

WORKED MODEL 09 | Delta to star

Question: Delta sides are RAB=6 Ω, RBC=9 Ω and RCA=3 Ω. Convert to star.

1. Sum of delta resistances: 6+9+3=18 Ω.
2. RA=(6×3)/18=1 Ω.
3. RB=(6×9)/18=3 Ω and RC=(9×3)/18=1.5 Ω.

> **RESULT Answer:** Star arms: RA=1 Ω, RB=3 Ω, RC=1.5 Ω.

WORKED MODEL 10 | Star to delta

Question: Star arms RA=2 Ω, RB=3 Ω, RC=6 Ω. Find delta side AB.

1. Use side opposite arm RC: RAB=RA+RB+(RA RB)/RC.
2. RAB=2+3+(2×3)/6.

> **RESULT Answer:** RAB=6 Ω. Other delta sides require the corresponding cyclic formula.

2.4 Wheatstone bridge and symmetry

A Wheatstone bridge is **balanced** when both midpoints have equal potential, so bridge-branch current is zero. For top-left/top-right/bottom-left/bottom-right arms chosen consistently, the ratio of resistances on one side equals that on the other: $R_1/R_2=R_3/R_4$. Check the circuit labels before applying a memorized ratio.

WORKED MODEL 11 | Balanced bridge

Question: Left branch comprises 2 Ω above 4 Ω; right branch comprises 3 Ω above unknown Rx. Determine Rx for zero current through the midpoint link.

1. Equal upper-to-lower divider ratios give 2/4=3/Rx.
2. Cross-multiply: 2Rx=12.

> **RESULT Answer:** Rx=6 Ω.

2.5 Open/short identification and equivalent resistance

- If an ideal conductor bypasses a resistor, the resistor has V=0; in a passive network it has I=0.
- A disconnected branch has I=0 but its end-to-end voltage can be nonzero.
- When finding equivalent resistance, remove the load, kill independent sources (ideal voltage source → short; ideal current source → open), but keep dependent sources active. For dependent-source networks apply a test source to calculate Rtest=Vtest/Itest.

> **EXAM TRAP** Do not combine resistors just because they look adjacent. In series, their shared node must have no other branches; in parallel, both ends must connect to the same pair of nodes.

03 Capacitors and electrostatic energy

3.1 Capacitance and its controlling factors

Capacitance describes charge stored per volt across two conductors. For a parallel-plate capacitor with negligible fringing, C=εA/d. A dielectric raises ε and hence capacitance; doubling plate area doubles C, whereas doubling separation halves C.

$C=Q/V = \varepsilon A/d$ | $\varepsilon=\varepsilon_0\varepsilon_r$ | $i=C\,dv/dt$ | $E=\frac{1}{2}CV^2=\frac{1}{2}QV=Q^2/(2C)$

The relation C=Q/V applies to a linear capacitor with fixed geometry and dielectric. In time-dependent circuits, i=C dv/dt follows when C is constant. If C changes mechanically with time, use i=d(Cv)/dt.

Capacitors	Same quantity	Combination
Parallel	Same voltage	$C_{eq}=C_1+C_2+\ldots$; Q branches ∝ C
Series	Same magnitude of charge (initially uncharged case)	$1/C_{eq}=1/C_1+1/C_2+\ldots$; V drops ∝ 1/C
2 in series	Q common	$C_{eq}=C_1C_2/(C_1+C_2)$

WORKED MODEL 12 | Series capacitor voltages

Question: 3 μF and 6 μF are placed in series on 18 V DC. Find charge and each voltage.

1. $C_{eq}=(3\times6)/(3+6)=2$ μF.
2. Series charge $Q=C_{eq}V=2\times18=36$ μC.
3. $V(3\mu F)=36/3=12$ V; $V(6\mu F)=36/6=6$ V.

RESULT Answer: 36 μC; 12 V and 6 V, respectively.

WORKED MODEL 13 | Parallel capacitor charge

Question: 2 μF and 4 μF are connected in parallel across 10 V.

1. Both have V=10 V.
2. $Q_2=2\times10=20$ μC; $Q_4=4\times10=40$ μC.
3. $C_{eq}=6$ μF; total Q=60 μC.

RESULT Answer: Total charge 60 μC, total energy $\frac{1}{2}(6\ \mu F)(10^2)=300$ μJ.

3.2 Inserting a dielectric: disconnected versus connected

Condition	Maintained	If dielectric fills all space and C → kC
Disconnected, isolated	Free charge Q fixed	V → V/k, energy → U/k
Connected to ideal DC source	Voltage V fixed	Q → kQ, stored energy → kU

At constant V, the energy stored increases even though electrostatic force tends to pull the dielectric in; the source supplies additional work. At constant Q, energy decreases as the dielectric enters.

WORKED MODEL 14 | Dielectric insertion in isolation

Question: An isolated 5 μF capacitor initially has 20 V. Relative permittivity 4 fills the gap.

1. Initially Q=5 μF×20 V=100 μC. Since isolated, Q is constant.
2. New capacitance=4×5=20 μF.
3. New voltage=Q/Cnew=100/20=5 V; final energy is one quarter of initial.

> **RESULT Answer:** Vnew=5 V; Q unchanged at 100 μC.

WORKED MODEL 15 | Energy released by charge sharing

Question: An initially charged 4 μF capacitor is at 12 V and a 2 μF capacitor is uncharged. Connect them in parallel with like-polarity plates joined; ignore wire resistance for final state.

1. Conserved total charge=4×12=48 μC.
2. Final common voltage=48/(4+2)=8 V.
3. Ui=½×4×12²=288 μJ; Uf=½×6×8²=192 μJ.

> **RESULT Answer:** Vfinal=8 V; 96 μJ is dissipated/radiated during redistribution.

3.3 Capacitor DC behavior and breakdown

- At DC steady state after switching effects have vanished, capacitor current is zero; capacitor voltage need not be zero.
- A capacitor charged to V cannot immediately change to a different voltage if capacitor current is finite; $v_C(0+)=v_C(0-)$.
- Breakdown voltage depends on dielectric strength and separation. Capacitance value alone does not determine safe operating voltage.

> **EXAM TRAP** Do not assume equal voltage across series capacitors. Series charge magnitudes are equal only under the standard initially uncharged ideal-series assumptions; precharged isolated internal nodes require charge-conservation analysis.

04 Inductors, coupling, batteries and source models

4.1 Inductance and magnetic energy

Self-inductance connects coil flux linkage $\lambda=N\Phi$ to current: $L=\lambda/I$ for a linear magnetic circuit. For a constant L, the induced terminal voltage in passive sign convention is $v=L\,di/dt$, and stored magnetic energy is $\frac{1}{2}LI^2$.

$\lambda=N\Phi$ | $L=N\Phi/I$ (linear) | $vL=L\,di/dt$ | $W_{mag}=\frac{1}{2}LI^2$

At DC steady state an ideal inductor is a short circuit ($vL=0$); a real coil has winding resistance and therefore a nonzero DC voltage drop. The current in an inductor cannot jump instantaneously unless an impulse of voltage is applied.

Connection	Equivalent inductance (uncoupled)
Series	$Leq=L_1+L_2+\ldots$
Parallel	$1/Leq=1/L_1+1/L_2+\ldots$
Coupled series aiding	$Leq=L_1+L_2+2M$
Coupled series opposing	$Leq=L_1+L_2-2M$

4.2 Mutual inductance and dot convention

$M=k\sqrt{(L_1L_2)}$, $0\le k\le 1$ | $v_1=L_1\,di_1/dt \pm M\,di_2/dt$

The plus/minus is set by reference current directions relative to dots. When both reference currents enter dotted terminals, mutual voltage terms have a consistent positive dot-to-undot sign. In series, aiding means fluxes reinforce (+2M); opposing means fluxes cancel (−2M).

WORKED MODEL 16 | Series-aiding and opposing coils

Question: Two coils have $L_1=4$ H, $L_2=9$ H, coefficient of coupling $k=0.5$. Find equivalent inductance for both series orientations.

1. $M=k\sqrt{(4\times9)}=0.5\times6=3$ H.
2. Aiding: $Leq=4+9+2\times3=19$ H.
3. Opposing: $Leq=4+9-2\times3=7$ H.

RESULT Answer: 19 H aiding; 7 H opposing.

WORKED MODEL 17 | Energy in an inductor

Question: A 0.2 H inductor carries 5 A. Find stored energy and the voltage when current rises at 40 A/s.

1. $W=\frac{1}{2}\times0.2\times5^2=2.5$ J.
2. $v=L(di/dt)=0.2\times40=8$ V (passive polarity).

RESULT Answer: Energy=2.5 J; voltage=8 V.

4.3 Batteries, EMF, terminal voltage

Model a discharging battery as ideal EMF E in series with internal resistance r. If current I is delivered to a load, terminal voltage $Vt=E-Ir$. During charging with current forced into the positive battery terminal, $Vt=E+Ir$ (with the corresponding current convention).

Discharging: $V_t = E - Ir$ | Short-circuit model current: $I_{sc} = E/r$ | Load: $I = E/(R+r)$

WORKED MODEL 18 | Practical cell and internal resistance

Question: A 12 V ideal EMF source has internal resistance 0.5 Ω and powers a 5.5 Ω load.

1. Total series resistance=5.5+0.5=6 Ω.
2. Load current=12/6=2 A.
3. Terminal voltage=I×5.5=11 V; internal loss=I^2r=2 W.

RESULT Answer: 2 A, terminal voltage 11 V, internal loss 2 W.

4.4 Source transformations

A real voltage source Vs in series with Rs is externally equivalent to a current source Is=Vs/Rs in parallel with the **same** Rs. The arrows and polarities must be matched. Source transformation preserves port V-I behavior, not internal power dissipation interpretations inside an idealized equivalent.

Vs series Rs ↔ Is=Vs/Rs parallel Rs

WORKED MODEL 19 | Voltage source to Norton

Question: A 24 V source has series resistance 6 Ω. Determine the Norton equivalent.

1. Current source Is=Vs/Rs=24/6=4 A.
2. Put a 6 Ω resistance in parallel with the 4 A current source.

RESULT Answer: Norton: 4 A in parallel with 6 Ω.

05 KCL, KVL, nodal and mesh analysis

5.1 Kirchhoff laws and reference directions

KCL follows conservation of electric charge at a node: sum of currents entering equals sum leaving. KVL follows conservation of energy for lumped circuits under the usual circuit assumptions: algebraic sum of voltage rises and drops around a loop is zero.

> **At a node: ΣIk=0 | Around a closed loop: ΣVk=0**

- Choose a current direction and voltage polarity freely; a negative answer means the actual direction is opposite to your assumption.
- Write resistor current from node a to b as (Va−Vb)/R; consistently choose either “all leaving” or “all entering” for KCL.
- When traversing a resistor in direction of current, voltage change is −IR; a source traversed from − to + gives +Vs.

WORKED MODEL 20 | One-loop KVL

Question: A 20 V source drives 2 Ω, 3 Ω and 5 Ω resistors in series. Find current and voltage across 3 Ω.

1. Choose loop current I clockwise. KVL: +20−2I−3I−5I=0.
2. Solve: 10I=20, so I=2 A.
3. Voltage across 3 Ω is 3×2=6 V.

> **RESULT Answer:** 2 A and 6 V.

5.2 Nodal analysis: general recipe

- Step 1: Select the reference (ground) node; name other node potentials V_1, V_2, … .
- Step 2: For each unknown node, write KCL: sum[(Vnode−Vneighbor)/R]=algebraic injected current.
- Step 3: Use voltage-source constraints when present; do not try to apply Ohm law across an ideal voltage source.
- Step 4: Solve the simultaneous equations and check signs, KCL and energy balance.

WORKED MODEL 21 | Two-node analysis

Question: Node A connects to 12 V through 2 Ω, to ground through 4 Ω, and to node B through 2 Ω. Node B connects to ground through 2 Ω. Find Va and Vb.

1. At A: (Va−12)/2+Va/4+(Va−Vb)/2=0. Multiply by 4: 5Va−2Vb=24.
2. At B: (Vb−Va)/2+Vb/2=0, giving Va=2Vb.
3. Substitute: 10Vb−2Vb=24 → Vb=3 V, Va=6 V.
4. Check: current from 12 V source into A=(12−6)/2=3 A; A-to-ground 1.5 A, A-to-B 1.5 A.

> **RESULT Answer:** Va=6 V, Vb=3 V.

5.3 Supernode method

If an ideal voltage source joins two non-reference nodes, enclose both nodes in one supernode. Write KCL for currents leaving the supernode to the external network, then write one voltage-source relation, such as Va−Vb=Vs, using the source polarity.

WORKED MODEL 22 | Supernode with a 5 V source

Question: Node A is connected to ground by 2 Ω, node B by 3 Ω. A 5 V source joins A (+) to B (−). A 5 A current source injects current from ground into A. Find node voltages.

1. Treat A and B as a supernode. Outgoing resistor currents must equal injected 5 A: $Va/2+Vb/3=5$.
2. Voltage-source constraint: $Va-Vb=5$.
3. Substitute $Va=Vb+5$: $(Vb+5)/2+Vb/3=5$. Multiply by 6: $3Vb+15+2Vb=30$.
4. Thus $Vb=3$ V and $Va=8$ V.

RESULT Answer: $Va=8$ V; $Vb=3$ V.

5.4 Mesh-current analysis

Assign a clockwise current to each independent planar mesh. In a shared resistor of value R, the voltage drop in mesh 1 is $R(I_1-I_2)$ if I_2 travels opposite across that branch. Write KVL for each mesh.

WORKED MODEL 23 | Two-mesh network

Question: Two clockwise mesh currents I_1 and I_2 share a 2 Ω resistor. Left exclusive resistor=4 Ω with 20 V source aiding I_1; right exclusive resistor=6 Ω with 10 V source aiding I_2.

1. Left mesh: $(4+2)I_1-2I_2=20 \rightarrow 6I_1-2I_2=20$.
2. Right mesh: $-2I_1+(6+2)I_2=10 \rightarrow -2I_1+8I_2=10$.
3. First gives $I_2=3I_1-10$. Substitute into second: $-2I_1+24I_1-80=10 \rightarrow 22I_1=90$.
4. $I_1=45/11$ A; $I_2=25/11$ A. Shared branch in I_1 direction: $I_1-I_2=20/11$ A.

RESULT Answer: $I_1\approx4.091$ A; $I_2\approx2.273$ A; shared branch≈1.818 A.

5.5 Supermesh and current sources

If an ideal current source lies between two adjacent meshes, form a supermesh that excludes that branch. Write KVL around the outer perimeter and add one constraint from the current source. With clockwise reference mesh currents, the shared-branch current is I_1-I_2 or I_2-I_1 depending on the indicated source arrow.

- If a current source belongs only to one boundary mesh, that mesh current is directly determined (signed) by the source.
- Dependent sources are not deactivated in circuit solutions. Add their controlling equations and solve simultaneously.

EXAM TRAP Node and mesh methods are alternatives. Node is often faster with current sources; mesh is often faster with voltage sources and a few planar loops.

5.6 Consistency checks: power balance and units

With passive sign convention, total delivered power equals total absorbed power. A computed 3 A through a 4 Ω resistor must dissipate $I^2R=36$ W. Negative computed resistor dissipation signals a sign or model error.

06 Network theorems and maximum power

6.1 Thevenin theorem: exact five-step procedure

- 1. Identify the load terminals a-b; remove the load. Mark polarity Va−Vb.
- 2. Compute open-circuit terminal voltage Voc; this is Vth.
- 3. Compute Rth looking into the port with independent sources killed. Keep dependent sources active; if they exist, drive the port with a test source and calculate Rth.
- 4. Replace the network by Vth in series with Rth. Reconnect RL.
- 5. Obtain IL=Vth/(Rth+RL), VL=IL RL, PL=IL²RL; check limiting cases.

Vth=Voc | Rth=Req at port with independent sources deactivated | IL=Vth/(Rth+RL)

WORKED MODEL 24 | Thevenin from voltage divider

Question: A 12 V source with 4 Ω series resistor feeds a 6 Ω shunt resistor to ground. Find Thevenin equivalent across the 6 Ω resistor terminals after treating the 6 Ω resistor as part of the network, for an additional load connected across those same terminals.

1. With the additional load removed, Voc=12×6/(4+6)=7.2 V.
2. Kill ideal 12 V source (short). The two resistors from output node to ground are parallel: Rth=4||6=24/10=2.4 Ω.
3. Therefore the additional load sees Vth=7.2 V in series with 2.4 Ω.

RESULT Answer: Vth=7.2 V; Rth=2.4 Ω.

WORKED MODEL 25 | Thevenin with disconnected load

Question: A 24 V ideal source is in series with 4 Ω. A load of 8 Ω is connected across output terminals. Find load power using Thevenin.

1. Remove load: no current in 4 Ω, so Voc=24 V.
2. Deactivate source: Rth=4 Ω.
3. Reconnect 8 Ω: I=24/(4+8)=2 A.
4. Power PL=I²RL=4×8=32 W.

RESULT Answer: 32 W.

6.2 Norton theorem and source conversion

Norton equivalent is a current source IN in parallel with RN=Rth. If the circuit is linear, IN is terminal short-circuit current, subject to modeling consistency. Thevenin and Norton are terminally equivalent: Vth=IN RN.

IN=Isc | RN=Rth | Vth=IN RN

WORKED MODEL 26 | Converting Norton to Thevenin

Question: A Norton source is 5 A in parallel with 3 Ω. Determine Thevenin form and voltage of a 6 Ω load.

1. Vth=IN RN=5×3=15 V.
2. Rth=RN=3 Ω.
3. Load voltage=15×6/(3+6)=10 V.

RESULT Answer: Thevenin 15 V in series with 3 Ω; VL=10 V.

6.3 Superposition theorem

In a **linear** circuit with multiple independent sources, the total voltage or current at a branch is the signed algebraic sum from each source acting alone. When turning off independent sources, short ideal voltage sources and open ideal current sources. Keep all resistances and dependent sources.

> **EXAM TRAP** You cannot add individual-source power values to get total load power because power is quadratic. First add signed voltages/currents, then compute total power.

WORKED MODEL 27 | Superposition with two supplies

Question: A 6 Ω resistor is connected between a node and ground. Two sources drive that node: 12 V through 3 Ω and 6 V through 6 Ω. Find voltage across the grounded 6 Ω resistor by superposition.

1. Activate 12 V source only; short 6 V source. Parallel branch from node to ground is 6||6=3 Ω. Node $V_1=12\times3/(3+3)=6$ V.
2. Activate 6 V source only; short 12 V source. Node-to-ground shunt is 3||6=2 Ω. Node $V_2=6\times2/(6+2)=1.5$ V.
3. Add contributions V=6+1.5=7.5 V.

> **RESULT Answer:** Load current=7.5/6=1.25 A. Do not add powers from source-by-source cases.

6.4 Maximum power transfer

For a positive Thevenin resistance Rth and a freely variable pure-resistor load, power $P_L=V_{th}^2R_L/(R_{th}+R_L)^2$. Differentiating with respect to RL gives the optimum RL=Rth.

> **DC resistive: RL=Rth; $P_{max}=V_{th}^2/(4R_{th})$; η at matching = 50%**

For sinusoidal AC with general complex source impedance Zth=Rth+jXth, unconstrained complex load has optimum $Z_L=Z_{th}^*=R_{th}-jX_{th}$ (conjugate match). Under a real-only load constraint, optimum RL=|Zth|, **not** Rth in general.

> **AC complex-load match: ZL=Zth* | $P_{max}=|V_{th,rms}|^2/(4R_{th})$ if Rth>0**

WORKED MODEL 28 | Maximum power transfer to a resistor

Question: An 18 V Thevenin source has Rth=3 Ω. Find optimum load resistance and power.

1. Set RL=Rth=3 Ω.
2. Current=18/(3+3)=3 A.
3. $P_L=I^2R_L=9\times3=27$ W.

> **RESULT Answer:** RL=3 Ω and Pmax=27 W.

WORKED MODEL 29 | AC complex matching

Question: A 20 V RMS Thevenin voltage source has Zth=4+j3 Ω. Find the complex impedance that maximizes load power.

1. Choose conjugate impedance ZL=4−j3 Ω.
2. Total impedance=8+j0 Ω. Source current=20/8=2.5 A RMS.
3. $P_L=I^2$ Re(ZL)=6.25×4=25 W.

> **RESULT Answer:** ZL=4−j3 Ω; maximum load power=25 W.

6.5 Special source and theorem traps

- With a dependent source, a test source is the safest way to find Rth; independently deactivating only the independent sources is necessary but does not fully solve the port resistance.

- A source current through an ideal short may be nonzero; do not calculate it as V/0.
- Thevenin/Norton descriptions are guaranteed at a specified port and for the operating assumptions; they do not tell you the internal distribution of current in the original network.

07 Magnetic circuits, materials and induction

7.1 Magnetic quantities and analogy with electricity

Magnetic quantity	Definition / unit	Analogous circuit idea
Magnetic flux Φ	Weber (Wb)	Current I
Flux density B	B=Φ/A, tesla (T)	Current density
Magnetic field intensity H	Ampere-turns/metre (A/m)	Electric field intensity
Permeability μ	μ=μ₀μr, H/m	Conductivity inverse relation
MMF F	F=NI, ampere-turn (AT)	Voltage V
Reluctance ℜ	ℜ=ℓ/(μA), AT/Wb	Resistance R

Φ=F/ℜ=NI/ℜ | B=Φ/A | H=NI/ℓ (uniform path) | B=μH

Reluctances add in series if the same flux passes through each section; for parallel flux paths, flux divides and reluctances combine by reciprocal sums in the ideal magnetic-circuit approximation. For different areas/permeabilities, find each section MMF drop Hiℓi and sum NI=ΣHiℓi.

7.2 Air gaps, leakage and fringing

Air has permeability approximately $\mu_0 = 4\pi \times 10^{-7}$ H/m. Even a short air gap may dominate total reluctance because the iron core usually has much greater permeability. Simple models neglect leakage flux and fringing; practical air-gap effective area may be slightly larger due to fringing.

ℜcore=ℓcore/(μcore Acore) | ℜgap=g/(μ₀ Agap) | NI=Φ(ℜcore+ℜgap)

WORKED MODEL 30 | Magnetic-circuit MMF

Question: A linear iron core has effective length 0.4 m, cross-sectional area 4 cm², μr=1000 and one air gap of 1 mm with the same area. Desired B=1 T. Find approximate ampere-turns, neglect fringing.

1. $H_{core} = B/(\mu_0\mu_r) = 1/(4\pi \times 10^{-7} \times 1000) \approx 795.8$ A/m. Core MMF≈795.8×0.4=318.3 AT.
2. $H_{gap} = B/\mu_0 \approx 795775$ A/m. Gap MMF≈795775×0.001=795.8 AT.
3. Total MMF≈318.3+795.8=1114.1 AT.

RESULT Answer: ≈1114 ampere-turns; for N=500 turns, I≈2.228 A.

7.3 Field formulas: wire, solenoid, toroid

Long straight wire: B=μI/(2πr) | Long solenoid: B≈μNI/ℓ | Toroid: H≈NI/(2πrmean)

For a circular loop of radius r, magnetic flux density at its centre is B=μNI/(2r). Use Ampere law for symmetric paths and Biot–Savart where geometry demands it. The long-solenoid result assumes field concentrated inside and negligible end effects.

WORKED MODEL 31 | Magnetic field near a conductor

Question: Calculate magnetic flux density at 2 cm from a long straight wire carrying 10 A in air.

1. Use $B=\mu_0 I/(2\pi r)$, $r=0.02$ m.
2. $B=(4\pi\times10^{-7}\times10)/(2\pi\times0.02)=1\times10^{-4}$ T.

RESULT Answer: 0.1 mT (100 μT).

7.4 Magnetic materials and B-H curve

- Diamagnetic: small negative susceptibility; paramagnetic: small positive susceptibility; ferromagnetic: large nonlinear permeability and hysteresis.
- Retentivity/remanence: B remaining when magnetizing H returns to zero after magnetization. Coercivity: reverse H required to reduce B to zero.
- Soft magnetic materials: narrow hysteresis loop, low coercivity; suitable for cores. Hard magnetic materials: wide loop, useful as permanent magnets.
- Hysteresis-loop area gives energy loss per unit volume per cycle (J/m³ per cycle); eddy current loss is reduced by laminating conducting cores.

7.5 Faraday law, Lenz law and motional EMF

$e=-N\,d\Phi/dt$ | $|e|=B\ell v$ (perpendicular motion) | Force on conductor: $F=B\,I\,\ell\,\sin\theta$

The minus sign in Faraday law expresses Lenz law: induced EMF opposes the change in flux linkage. For a coil, terminal reference convention and winding orientation decide measured polarity; magnitudes are usually sufficient for basic numerical questions.

WORKED MODEL 32 | Sinusoidal flux and induced EMF

Question: A 200-turn coil has $\Phi(t)=0.01\sin(100\pi t)$ Wb per turn. Find maximum induced EMF.

1. Flux linkage $\lambda=N\Phi(t)=2\sin(100\pi t)$ Wb-turn.
2. Magnitude of induced EMF$=|d\lambda/dt|=200\pi|\cos(100\pi t)|$ V.
3. Peak EMF$=200\pi$ V≈628.3 V.

RESULT Answer: Approximately 628 V.

WORKED MODEL 33 | Motional EMF

Question: A straight 0.5 m rod moves at 8 m/s perpendicular to a uniform 0.4 T field; velocity and rod are mutually perpendicular.

1. Use magnitude $e=B\ell v$.
2. $e=0.4\times0.5\times8$.

RESULT Answer: 1.6 V.

7.6 Self-inductance from magnetic geometry

For a long solenoid of N turns, length ℓ, area A and linear permeability μ, approximately $L=\mu N^2A/\ell$. For a gapped high-permeability magnetic circuit, $L\approx N^2/\mathfrak{R}$total, within a linear, no-leakage model.

$L\approx\mu N^2A/\ell$ | $L\approx N^2/\mathfrak{R}$total | Energy density $=B^2/(2\mu)$ (linear medium)

EXAM TRAP Iron permeability is not generally constant. If an examination supplies a nonlinear B-H curve, obtain H from the supplied B value rather than blindly applying one fixed μr.

08 AC waveforms, RMS, average values and phasors

8.1 Sinusoidal waveform and angular relationships

$v(t)=V_m \sin(\omega t+\theta)$ | $\omega=2\pi f$ | $T=1/f$ | Electrical phase angle $=\omega\Delta t$

Vm is peak voltage. For a sinusoidal waveform, peak-to-peak magnitude is 2Vm. Phase angle between two waves equals the difference of their angle constants when the same trigonometric reference (sin or cos) is used. A positive phase relative to the common reference means a waveform reaches corresponding points earlier in time (leads).

WORKED MODEL 34 | Frequency, period and phase

Question: A sinusoid of frequency 50 Hz has a time displacement of 2 ms relative to another sinusoid.

1. $T=1/50=20$ ms.
2. Phase $=360°\times(2/20)=36°$.

RESULT Answer: 36° phase displacement. Time ordering determines which signal leads.

8.2 RMS definition and derivation

The RMS of a periodic voltage is the DC voltage that produces the same average resistive heating in the same resistor. For period T, $V_{rms}=\sqrt{[(1/T)\int_0^T v^2(t)dt]}$. For sine $v=V_m \sin \omega t$, mean$(\sin^2)=1/2$, so $V_{rms}=V_m/\sqrt{2}$. This identity depends on the waveform shape, not just its peak.

$V_{rms}=\sqrt{[(1/T)\int_0^T v^2 dt]}$ | Sine $V_{rms}=V_m/\sqrt{2}\approx 0.707V_m$

8.3 Average value and waveform factors

A symmetrical AC sine wave averages to **zero over a complete cycle**. Its average over the positive half-cycle, or the mean of its full-wave rectified magnitude, is $2V_m/\pi$. Questions often omit the averaging interval; check whether they mean full-cycle average or rectified average.

Full-cycle sine average $=0$ | Positive-half mean $=2V_m/\pi\approx 0.637V_m$

Sine form factor $=V_{rms}/V_{rect\text{-}avg}=\pi/(2\sqrt{2})\approx 1.11$ | Crest factor $=V_m/V_{rms}=\sqrt{2}\approx 1.414$

Waveform (peak=Vm)	RMS	Full-period average	Average |v|
Sinusoid	$V_m/\sqrt{2}$	0	$2V_m/\pi$
Symmetric square, ±Vm	Vm	0	Vm
Symmetric triangular, ±Vm	$V_m/\sqrt{3}$	0	Vm/2
Unipolar half-wave sine	Vm/2	V_m/π	V_m/π

WORKED MODEL 35 | RMS and peak of sine

Question: A sinusoidal voltage has 230 V RMS. Find its peak, peak-to-peak and rectified average.

1. $V_m=\sqrt{2}\times230\approx325.3$ V.
2. Peak-to-peak $=2V_m\approx650.5$ V.
3. Full-wave-rectified mean $=(2/\pi)V_m\approx207.1$ V.

RESULT Answer: Peak≈325.3 V; p-p≈650.5 V; rectified mean≈207.1 V.

WORKED MODEL 36 | RMS of unequal-step waveform

Question: A periodic voltage equals 10 V for half its period and 0 V for the other half.

1. $V_{rms}^2=(10^2\times0.5)+(0^2\times0.5)=50$.
2. $V_{rms}=\sqrt{50}=10/\sqrt{2}\approx7.071$ V.
3. Arithmetic average $=10\times0.5=5$ V.

RESULT Answer: RMS≈7.071 V; average=5 V.

8.4 Phasor basics and complex arithmetic

A sinusoidal quantity $x(t)=\sqrt{2}\,X\sin(\omega t+\theta)$ corresponds to an RMS phasor $X\angle\theta$ under a sine reference. In phasor circuit equations use $j=\sqrt{-1}$ and exponential representation $X\angle\theta=X(\cos\theta+j\sin\theta)$.

$j^2=-1$ | jX rotates a phasor by +90° | $1/j=-j$ | $|a+jb|=\sqrt{(a^2+b^2)}$

Phasors add as rectangular complex numbers, not by adding magnitudes. For example $3\angle0°+4\angle90°=3+j4=5\angle53.13°$. Use the same angular frequency for all phasors in a single steady-state analysis.

WORKED MODEL 37 | Phasor addition

Question: Find the magnitude and angle of $I=6\angle0°+8\angle90°$ A.

1. Convert rectangular: $I=6+j8$ A.
2. Magnitude $=\sqrt{(6^2+8^2)}=10$ A.
3. Angle $=\tan^{-1}(8/6)=53.13°$.

RESULT Answer: $10\angle53.13°$ A.

8.5 Lead/lag quick reference

Element under sinusoidal steady state	Current versus voltage	Power factor
Pure resistor	In phase (0°)	Unity
Pure inductor	Current lags voltage 90°	Zero lagging
Pure capacitor	Current leads voltage 90°	Zero leading
RL load	Current lags by 0° to 90°	Lagging
RC load	Current leads by 0° to 90°	Leading

EXAM TRAP Power factor is cos of the phase angle between voltage and current at the same port. It is not generally R/Z if Z is a complex symbol; use R/|Z| for a series RL or RC circuit.

09 AC circuits, impedance, power and correction

9.1 R, L and C impedance

In sinusoidal steady state, a linear resistor has impedance R; an ideal inductor jωL and an ideal capacitor 1/(jωC)=−j/(ωC). Impedance Z=V/I; admittance Y=1/Z=G+jB. Complex impedance accounts for phase as well as magnitude.

$Z_R=R$ | $Z_L=jX_L=j\omega L$ | $Z_C=-jX_C=1/(j\omega C)$ | $X_L=2\pi fL$ | $X_C=1/(2\pi fC)$

Component	Magnitude	Current phase vs V	Effect of increasing f
R	R	0°	Unchanged for ideal R
L	XL=ωL	−90°	Reactance increases
C	XC=1/(ωC)	+90°	Reactance decreases

Physical intuition: an ideal inductor opposes *change in current*; at higher sinusoidal frequency it has greater reactance. A capacitor opposes *change in voltage*; at higher frequency it passes more alternating current because its reactance falls.

WORKED MODEL 38 | Reactance at a stated frequency

Question: Calculate reactance of L=0.1 H and C=100 μF at f=50 Hz.

1. $X_L=2\pi fL=2\pi\times50\times0.1=10\pi\approx31.42\ \Omega$.
2. $X_C=1/(2\pi fC)=1/(2\pi\times50\times100\times10^{-6})=100/\pi\approx31.83\ \Omega$.

RESULT Answer: $X_L\approx31.42\ \Omega$, $X_C\approx31.83\ \Omega$.

9.2 Series R-L, R-C, R-L-C circuits

$Z_{series}=R+j(X_L-X_C)$ | $|Z|=\sqrt{[R^2+(X_L-X_C)^2]}$ | $\varphi=\tan^{-1}[(X_L-X_C)/R]$

For a series RLC circuit, I=V/|Z|; resistor voltage VR=IR; inductor VL=IXL; capacitor VC=IXC. The source voltage is the **phasor** sum: $V^2=V_R^2+(V_L-V_C)^2$ when all elements are series and source RMS voltage is used. Adding |VR|+|VL|+|VC| ordinarily gives the wrong source magnitude.

WORKED MODEL 39 | RL circuit and power factor

Question: A series load R=6 Ω and XL=8 Ω is connected to 100 V RMS.

1. $Z=6+j8\ \Omega$; $|Z|=\sqrt{(36+64)}=10\ \Omega$.
2. $I=100/10=10$ A.
3. Power factor $\cos\varphi=R/|Z|=6/10=0.6$ lagging.
4. $P=I^2R=100\times6=600$ W; $Q=I^2X_L=100\times8=800$ var.

RESULT Answer: I=10 A; PF=0.6 lagging; P=600 W; Q=800 var.

WORKED MODEL 40 | Series RC circuit

Question: A 3 Ω resistor and capacitor reactance 4 Ω are in series on 50 V RMS.

1. $Z=3-j4\ \Omega$; $|Z|=5\ \Omega$.
2. $I=50/5=10$ A.
3. $\varphi=-\tan^{-1}(4/3)=-53.13°$ impedance angle, so current leads the applied voltage by 53.13°.
4. $PF=R/|Z|=0.6$ leading; $P=I^2R=300$ W.

> **RESULT Answer:** 10 A; 0.6 leading power factor; 300 W.

WORKED MODEL 41 | Series RLC with partial cancellation

Question: R=12 Ω, XL=25 Ω and XC=16 Ω at a fixed frequency; applied voltage=150 V RMS.

1. Net reactance $X=25-16=+9\ \Omega$; inductive.
2. $|Z|=\sqrt{(12^2+9^2)}=15\ \Omega$.
3. $I=150/15=10$ A; $PF=12/15=0.8$ lagging.
4. VR=120 V, VL=250 V, VC=160 V; $\text{sqrt}(120^2+(250-160)^2)=150$ V.

> **RESULT Answer:** I=10 A, PF=0.8 lagging, resistor P=1200 W.

9.3 Parallel AC circuits and admittance

Parallel branches share applied voltage, while branch currents add **vectorially**. For an RL branch, calculate branch current as V/(R+jXL). For a pure R and a pure L in parallel, resistor current is real (in phase), inductor current is negative imaginary relative to V. Use total $I=\Sigma Ik$ or $Yeq=\Sigma(1/Zk)$.

> **$Yeq=1/Zeq=\Sigma(1/Zk)$ | $I=VYeq$ | For parallel R and L: $|I|=\sqrt{[(V/R)^2+(V/XL)^2]}$**

WORKED MODEL 42 | Parallel resistor and inductor

Question: A 10 Ω resistor and a pure inductor with XL=10 Ω are in parallel on 100 V RMS.

1. Resistor branch current=$100/10=10\angle 0°$ A.
2. Inductor branch current=$100/10=10\angle -90°=-j10$ A.
3. Source $I=10-j10=14.142\angle -45°$ A.
4. Total $PF=\cos 45°=0.7071$ lagging; $P=V^2/R=1000$ W.

> **RESULT Answer:** Source current 14.14 A; PF≈0.707 lagging.

WORKED MODEL 43 | Parallel RC with leading current

Question: A 20 Ω resistor and capacitor reactance XC=15 Ω are in parallel on 60 V RMS.

1. $IR=60/20=3$ A in phase.
2. $IC=60/15=4$ A, 90° leading → $+j4$ A.
3. $I=3+j4=5$ A at +53.13°.
4. $PF=3/5=0.6$ leading; $P=60\times 3=180$ W.

> **RESULT Answer:** I=5 A, PF=0.6 leading, P=180 W.

9.4 Real, reactive, apparent and complex power

For a sinusoidal steady-state port, define $\varphi=\theta V-\theta I$. Active power P measures net average energy transfer; reactive Q measures reversible exchange with fields; apparent power |S| combines them. Use **RMS phasors** for complex power.

$S=V I^* = P+jQ$ | $P=VI\cos\varphi$ | $Q=VI\sin\varphi$ | $|S|=VI=\sqrt{(P^2+Q^2)}$

For any impedance $Z=R+jX$: $P=I^2R$, $Q=I^2X$ | $pf=P/|S|=\cos\varphi$ (sinusoidal)

Inductive (lagging) loads: $Q>0$; capacitive (leading) loads: $Q<0$. A purely reactive ideal element consumes zero average real power. For nonsinusoidal waveforms, true power factor need not equal displacement $\cos\varphi$ because harmonic distortion also matters.

Power measure	Symbol	Unit
Real/active power	P	watt (W)
Reactive power	Q	volt-ampere reactive (var)
Apparent power	|S|	volt-ampere (VA)
Complex power	S=P+jQ	VA (complex)

WORKED MODEL 44 | Power triangle model

Question: A single-phase load draws 10 kVA at 0.8 lagging PF. Find P and Q.

1. $P=S\cos\varphi=10\times0.8=8$ kW.
2. $\sin\varphi=\sqrt{(1-0.8^2)}=0.6$.
3. $Q=S\sin\varphi=10\times0.6=6$ kvar, positive for inductive.

RESULT Answer: P=8 kW; Q=+6 kvar.

WORKED MODEL 45 | Power using complex impedance

Question: A load has 230 V RMS and current $10\angle-30°$ A with $V=230\angle0°$ V.

1. Conjugate current $I^*=10\angle+30°$.
2. Complex power $S=230\times10\angle30°=2300(\cos30°+j\sin30°)$.
3. $P=2300\times0.866\approx1991.9$ W; $Q=2300\times0.5=1150$ var.

RESULT Answer: P≈1.992 kW, Q=1.15 kvar; PF=0.866 lagging.

9.5 Power-factor correction with shunt capacitor

An ideal shunt capacitor supplies negative reactive power, reducing net positive reactive power drawn from the source while keeping real load power approximately constant. Starting with original lagging PF $\cos\varphi_1$ and target PF $\cos\varphi_2$, required capacitor var magnitude is $P(\tan\varphi_1-\tan\varphi_2)$.

$Q_c = P(\tan\varphi_1-\tan\varphi_2)$ | 1-phase: $C = Q_c/(\omega V^2)$

For a 3-phase shunt capacitor bank at line voltage VL, star: $Q_c=\omega(3\ C_Y V_{ph}^2)=\omega\ C_Y V_L^2$. Delta: $Q_c=3\omega\ C_\Delta V_L^2$. Here C is **per phase**. Choose star or delta based on the actual capacitor-bank connection.

WORKED MODEL 46 | Single-phase PF improvement

Question: An 8 kW single-phase load at 0.6 lagging PF is corrected to 0.8 lagging at 230 V, 50 Hz. Find capacitor size.

1. $\tan\varphi_1 = 0.8/0.6 = 4/3$; $\tan\varphi_2 = 0.6/0.8 = 3/4$.
2. $Q_c = 8000[(4/3) - (3/4)] = 8000 \times 7/12 = 4666.7$ var.
3. $C = Q_c/(2\pi \times 50 \times 230^2) \approx 2.808 \times 10^{-4}$ F.

> **RESULT Answer:** Approximately 281 µF shunt capacitance.

WORKED MODEL 47 | Corrected supply current

Question: A 10 kW load at 200 V single phase has PF 0.5 lagging; ideal correction raises PF to unity.

1. Before correction: $I = P/(V \text{pf}) = 10000/(200 \times 0.5) = 100$ A.
2. After correction: $I = P/(V \times 1) = 10000/200 = 50$ A.

> **RESULT Answer:** Supply current halves to 50 A (load real power unchanged).

9.6 Exam-safe procedure for any AC circuit

- Convert units: mH→H; µF→F; voltage/current to RMS unless told otherwise.
- Compute $\omega = 2\pi f$, $X_L = \omega L$ and $X_C = 1/(\omega C)$; assign signs $+jX_L$ and $-jX_C$.
- Reduce series impedances by addition; parallel impedances using admittance or product/sum.
- Find total $I = V/Z$ in complex form, then its magnitude and angle.
- Compute P, Q, $|S|$ and PF from $V I^*$; check if lagging/leading agrees with net reactance sign.

> **EXAM TRAP** When AC frequency changes, X_L changes directly and X_C inversely. **Do not** reuse 50 Hz reactances unchanged at 60 Hz.

10 Resonance, bandwidth and tank circuits

10.1 Series RLC resonance

Series resonance occurs when XL=XC, so net reactance cancels, impedance is purely R, current is maximum for a fixed applied sinusoidal voltage, and source power factor is unity. Individual inductor and capacitor voltages may be much greater than applied voltage; they are opposite phasors and cancel in the sum.

$\omega_0=1/\sqrt{(LC)}$ | $f_0=1/(2\pi\sqrt{LC})$ | Z at resonance=R | Imax=V/R

At resonant frequency, quality factor $Qs=\omega_0 L/R=1/(\omega_0 CR)$. For series RLC, the exact half-power bandwidth in angular frequency is $\Delta\omega=R/L$ and in hertz $BW=R/(2\pi L)=f_0/Qs$. Lower/upper half-power frequencies satisfy $|XL-XC|=R$.

Series $Qs=\omega_0 L/R=1/(\omega_0 CR)$ | $BW=f_2-f_1=R/(2\pi L)$ | $Qs=f_0/BW$

At resonance: VL=VC=Q·Vsource (magnitudes) | Half-power current = Imax/√2

WORKED MODEL 48 | Resonant frequency

Question: L=0.1 H and C=100 μF form an ideal series RLC circuit. Find f_0.

1. $LC=0.1\times100\times10^{-6}=10^{-5}$.
2. $f_0=1/(2\pi\sqrt{10^{-5}})\approx50.33$ Hz.

RESULT Answer: Resonance at approximately 50.33 Hz.

WORKED MODEL 49 | Resonant current, Q and bandwidth

Question: A series RLC circuit has R=10 Ω, L=0.1 H and C=100 μF. The applied RMS voltage is 100 V.

1. Resonant frequency $f_0\approx50.33$ Hz from LC.
2. At resonance $I=V/R=100/10=10$ A.
3. $Qs=\omega_0 L/R=(316.23\times0.1)/10\approx3.162$.
4. Bandwidth $BW=R/(2\pi L)=10/(2\pi\times0.1)\approx15.92$ Hz.

RESULT Answer: Ires=10 A, Q≈3.162, BW≈15.92 Hz.

WORKED MODEL 50 | Voltage magnification at resonance

Question: A series-resonant circuit has Q=20 and source voltage 10 V RMS. Find individual inductor and capacitor voltage magnitudes.

1. At resonance magnification factor is Q.
2. VL=Q V=20×10=200 V; VC also 200 V, opposite in phase.
3. Source still sees only 10 V because reactive voltages cancel vectorially.

RESULT Answer: VL=VC=200 V RMS.

10.2 Exact half-power frequencies

At a series RLC half-power point, impedance magnitude increases from R to √2R; therefore $|\omega L-1/(\omega C)|=R$. Its two positive solutions satisfy $\omega_2-\omega_1=R/L$ and $\omega_1\omega_2=1/(LC)=\omega_0^2$. They are not generally symmetric around f_0, although they are approximately so for high-Q circuits.

$\omega_1=[\sqrt{(R^2+4L/C)}-R]/(2L)$ | $\omega_2=[\sqrt{(R^2+4L/C)}+R]/(2L)$

WORKED MODEL 51 | Half-power reasoning

Question: At series resonance, 10 A flows through a 5 Ω resistance. Find current and resistor power at a half-power frequency.

1. At resonance resistor power $P_{res}=I^2R=10^2\times5=500$ W.
2. At either half-power frequency $P=250$ W.
3. $I_{half}=\sqrt{(250/5)}=\sqrt{50}=10/\sqrt{2}$ A.

RESULT Answer: $I_{half}\approx7.071$ A; half-power resistor dissipation=250 W.

10.3 Parallel resonance and ideal tank circuit

For ideal L and C connected in parallel, total susceptance is $B=\omega C-1/(\omega L)$. At resonance $B=0$ and their branch reactive currents cancel. The equivalent ideal parallel LC presents infinite impedance (zero input current), even though circulating L and C branch currents are nonzero when a finite AC voltage is applied. Real parallel tanks have losses, giving a large but finite impedance and finite bandwidth.

Ideal parallel LC: $\omega_0=1/\sqrt{(LC)}$, $f_0=1/(2\pi\sqrt{LC})$, $|Z_{in}|\to\infty$ at resonance

For an ideal parallel RLC with shunt resistor Rp, $Y=1/R_p+j(\omega C-1/(\omega L))$. At resonance $Z_{in}=R_p$, $Q_p=R_p/(\omega_0 L)=\omega_0 C R_p$ and $BW=1/(2\pi R_p C)$ under this ideal parallel-RLC model.

Ideal shunt RLC: $Q_p=\omega_0 C R_p=R_p/(\omega_0 L)$ | $BW=1/(2\pi R_p C)$

WORKED MODEL 52 | Parallel tank

Question: A 10 mH inductor and 10 μF capacitor are in parallel with ideal components. Determine resonance frequency.

1. $LC=0.01\times10\times10^{-6}=10^{-7}$.
2. $f_0=1/(2\pi\sqrt{10^{-7}})\approx503.3$ Hz.

RESULT Answer: Ideal tank resonates at approximately 503.3 Hz.

10.4 High-frequency / low-frequency behavior and Q

Series RLC	Below resonance	At resonance	Above resonance
Net behavior	Capacitive	Resistive	Inductive
Circuit impedance magnitude	Above R	Minimum R	Above R
Current magnitude at fixed voltage	Less than max	Maximum	Less than max
Power factor	Leading	Unity	Lagging

EXAM TRAP Resonance formulas depend on the **topology and loss model**. For a practical coil with resistance in series with L, placed in parallel with C, the resonant frequency can differ from $1/(2\pi\sqrt{LC})$. Read the specified circuit.

11 First-order RC and RL switching response (extension)

This chapter is supporting practice. It is valuable for understanding actual element behavior and older exam-style circuit questions, though 2026 core wording does not separately list time-domain transients.

11.1 Universal first-order step response

If a linear first-order circuit has an initial value x(0+) and final value x(∞), then under stable constant-coefficient DC switching: x(t)=x(∞)+[x(0+)−x(∞)] e^(−t/τ). The time constant τ is positive and depends on the resistance seen by the storage element.

x(t)=xf+(x0−xf)e^(−t/τ) | RC: τ=Req C | RL: τ=L/Req

To find Req, deactivate independent sources and look into capacitor or inductor terminals in the post-switch configuration, respecting dependent sources. More complex circuits with multiple independent energy-storage states may not be first order.

Elapsed time	Fraction of initial deviation remaining	Fraction of final step completed
1τ	$e^{-1}\approx36.8\%$	63.2%
2τ	$e^{-2}\approx13.5\%$	86.5%
3τ	$e^{-3}\approx5.0\%$	95.0%
5τ	$e^{-5}\approx0.67\%$	99.33%

11.2 RC charging and discharging

Charging from 0 to Vs: vC=Vs(1−e^(−t/RC)), i=(Vs/R)e^(−t/RC)

Discharge from V0 through R: vC=V0 e^(−t/RC), i magnitude=(V0/R)e^(−t/RC)

WORKED MODEL 53 | RC capacitor charging

Question: A 100 μF initially uncharged capacitor charges from a 10 V ideal source via a 10 kΩ resistor. Find voltage after one time constant.

1. $\tau=RC=(100\times10^{-6})(10\times10^{3})=1$ s.
2. $vC(\tau)=10(1-e^{-1})\approx6.321$ V.
3. Initial charging current $i(0+)=Vs/R=10/10000=1$ mA.

RESULT Answer: At 1 s, capacitor voltage≈6.32 V.

WORKED MODEL 54 | RC half-voltage decay

Question: A capacitor is initially at V0 and discharges through R. Find time required to reach V0/2.

1. $V/V0=e^{(-t/RC)}=1/2$.
2. Take natural logarithms: $-t/RC=-\ln2$.

RESULT Answer: t=RC ln2≈0.693RC.

11.3 RL current growth and decay

RL energization from zero: i=Vs/R ·(1−e^(−tR/L)) | RL free decay: i=I0 e^(−tR/L)

WORKED MODEL 55 | RL energization

Question: A 2 H inductor and 4 Ω resistor are energized in series by 20 V DC. Find steady current, τ and current at τ.

1. Final current If=V/R=20/4=5 A.

2. τ=L/R=2/4=0.5 s.

3. $i(\tau)=5(1-e^{-1})\approx 3.161$ A.

RESULT Answer: Ifinal=5 A; τ=0.5 s; i(0.5s)≈3.16 A.

11.4 Immediately after switching: critical rules

- Capacitor voltage continuity: vC(0+)=vC(0−), absent an impulse current.
- Inductor current continuity: iL(0+)=iL(0−), absent an impulse voltage.
- A previously uncharged capacitor looks like a zero-volt source **at the instant 0+**, not necessarily like an open circuit.
- An initially zero-current inductor behaves as a zero-current source **at 0+**, not necessarily like a short circuit.
- At DC t→∞, ideal capacitor is open; ideal inductor short. Do not interchange initial and final circuit models.

12 Balanced three-phase circuits and star–delta loads

12.1 Three-phase generation and balance

A balanced three-phase set has equal phase RMS magnitudes, equal frequency and 120° phase separation. For a positive-sequence set using a cosine reference: VAN=Vph∠0°, VBN=Vph∠−120°, VCN=Vph∠+120°. Their phasor sum is zero.

A balanced three-phase load has the same impedance in each phase; phase currents have equal magnitude and are displaced by 120°. For a balanced load, neutral current is zero in a four-wire star circuit.

12.2 Star (Y) connection

In balanced star, each impedance lies between a line conductor and neutral. Phase current equals line current. Line-to-line voltage is the phasor difference of phase-to-neutral voltages and has magnitude √3 times phase voltage.

> **Balanced Y: VL=√3 Vph | IL=Iph | Vph=VL/√3**

For positive sequence, VAB=√3 VAN∠+30°. Do not confuse the 30° shift of line voltages with the load PF angle between phase voltage and phase current.

WORKED MODEL 56 | Star load line current

Question: A balanced star load has impedance 8+j6 Ω per phase and line voltage 400 V RMS. Find line current and PF.

1. Per-phase impedance magnitude |Z|=10 Ω; PF=8/10=0.8 lagging.
2. Vph=VL/√3=400/√3≈230.94 V.
3. Iph=Vph/|Z|=400/(10√3)≈23.09 A.
4. For star, IL=Iph.

> **RESULT Answer:** IL≈23.09 A at 0.8 lagging PF.

12.3 Delta (Δ) connection

In balanced delta, each impedance connects directly between two line terminals. Phase voltage equals line voltage. Line current is the phasor difference of two phase currents and has magnitude √3 times phase current.

> **Balanced Δ: VL=Vph | IL=√3 Iph | Iph=VL/|ZΔ|**

For the common positive-phase-sequence current convention, corresponding line current lags the associated delta branch current by 30°. The exact sign depends on line and branch reference directions, so write the current differences explicitly in phase-angle questions.

WORKED MODEL 57 | Delta load line current

Question: A balanced delta impedance is 12+j16 Ω per phase, connected to 400 V line-to-line supply.

1. |Zph|=√(12²+16²)=20 Ω; PF=12/20=0.6 lagging.
2. Each delta phase sees Vph=VL=400 V.
3. Iph=400/20=20 A.
4. Line current IL=√3×20≈34.64 A.

> **RESULT Answer:** IL≈34.64 A, PF=0.6 lagging.

12.4 Three-phase active, reactive and apparent power

Balanced 3φ: $P=\sqrt{3}\,V_L I_L \cos\varphi = 3V_{ph} I_{ph} \cos\varphi$

$Q=\sqrt{3}\,V_L I_L \sin\varphi$ | $|S|=\sqrt{3}\,V_L I_L$ | $S=P+jQ$

The same total-power expressions apply for balanced star and delta loads. **Only use φ for the angle between voltage and current in the same phase impedance**, not the 30° line-to-phase geometrical offset.

WORKED MODEL 58 | 3-phase power and reactive power

Question: A balanced 400 V, 3-phase load takes 20 A line current at 0.8 lagging PF.

1. Apparent power $|S|=\sqrt{3}\times400\times20\approx13.856$ kVA.
2. $P=|S|\times0.8\approx11.085$ kW.
3. $\sin\varphi=0.6$; $Q=|S|\times0.6\approx8.314$ kvar.

RESULT Answer: P≈11.09 kW; Q≈8.31 kvar.

WORKED MODEL 59 | Compute line current from 3-phase power

Question: A balanced motor draws 15 kW electrical input from 415 V 3-phase supply at 0.75 lagging PF.

1. $P=\sqrt{3}\,V_L I_L \cos\varphi$.
2. $I_L=15000/(\sqrt{3}\times415\times0.75)$.

RESULT Answer: IL≈27.82 A. Mechanical efficiency would be needed to calculate shaft output.

12.5 Star–delta impedance transformations

For a balanced three-terminal set, **equivalent load** conversion gives $Z_\Delta=3Z_Y$ and $Z_Y=Z_\Delta/3$. These describe the same terminal behavior under all balanced three-phase applied voltages, not merely identical nameplate connections. The equivalent delta impedances are three times their star counterparts.

Balanced equivalent conversion: $Z_\Delta=3\,Z_Y$ | $Z_Y=Z_\Delta/3$

WORKED MODEL 60 | Equivalent star–delta balanced load

Question: A balanced star-connected load has phase impedance 10+j5 Ω. Find the equivalent balanced delta impedance.

1. Apply $Z_\Delta=3Z_Y$.
2. $Z_\Delta=3(10+j5)=30+j15$ Ω per delta branch.

RESULT Answer: $Z_\Delta=30+j15$ Ω per phase (terminally equivalent).

12.6 Resistor heating and loads rated by line/phase voltage

When three identical resistor elements, each R, are connected in star at fixed VL, total $P_{star}=V_L^2/R$. Reconnecting those same physical resistors in delta at the same VL gives $P_{delta}=3V_L^2/R$, so $P_{delta}/P_{star}=3$. This is **not** the same as a terminally equivalent star-delta impedance conversion, where impedances change by factor 3 and total power is unchanged.

WORKED MODEL 61 | Same heaters reconnected

Question: Three 30 Ω heaters are connected to a 300 V three-phase supply first in star then in delta. Compare total power.

1. Star: $P_{star}=V_L^2/R=300^2/30=3000$ W.
2. Delta: $P_{delta}=3V_L^2/R=3\times300^2/30=9000$ W.
3. Ratio $P_{delta}/P_{star}=3$.

> **RESULT Answer:** Star 3 kW; delta 9 kW.

12.7 Neutral current and unbalanced cases

For a balanced star load, $I_A+I_B+I_C=0$ vectorially. If a four-wire star load is unbalanced and neutral is connected, calculate each phase current independently from its phase-to-neutral source voltage and impedance; neutral current is $-(I_A+I_B+I_C)$. With no neutral and an unbalanced star, the star point shifts, so assuming fixed $V_L/\sqrt{3}$ across each phase is generally wrong.

> **EXAM TRAP** √3 factors relate line and phase **magnitudes** only in balanced three-phase systems. Check star versus delta first, then check whether the question asks line current, branch current or total power.

13 Calculation models and exam decision tree

13.1 What to do first when you see a circuit

Question clue	First method	Why
All resistors in a reducible network	Series/parallel, short/open, star-delta	Avoid simultaneous equations
Unknown one node voltage	KCL / nodal	Few unknowns
Two adjacent loops with voltage sources	Mesh / KVL	Few unknown loop currents
Load changes, requested load current/power	Thevenin or Norton	Solve source network only once
Multiple independent sources	Superposition	Linear branch V/I contributions
Circuit at 50 Hz	Impedance/admittance	Reactance depends on frequency
Max load power	Match RL or ZL	Watch DC vs AC conditions
Magnetic core with air gap	MMF sum over sections	Gap often dominates
3-phase load	Identify star or delta	Line/phase relation differs

13.2 Twelve frequent question models with one-line tricks

Pattern	Fast relationship
Stretched wire	R scales as square of length factor if volume constant
Wire cut into n equal parallel parts	$R_{eq}=R/n^2$
Two equal resistors in parallel	$R_{eq}=R/2$
Two equal capacitors in series	$C_{eq}=C/2$
Peak sine voltage from RMS	$V_m=\sqrt{2}\,V_{rms}$
50 Hz waveform period	20 ms
Three-phase star voltage	$V_{ph}=V_L/\sqrt{3}$
Three-phase delta current	$I_L=\sqrt{3}\,I_{ph}$
Series resonance	$X_L=X_C$, source PF=1
Single-phase PF correction	$Q_c=P(\tan\varphi_1-\tan\varphi_2)$
Ideal inductor at DC steady state	Short circuit
Ideal capacitor at DC steady state	Open circuit

13.3 Mixed worked models

WORKED MODEL 62 | Bridge + resistance recognition

Question: A 5 Ω and 10 Ω resistor form a series branch, which is in parallel with 15 Ω. Find equivalent resistance.

1. First combine left branch: 5+10=15 Ω.
2. Now two 15 Ω branches in parallel: Req=15/2 Ω.

> **RESULT Answer:** 7.5 Ω.

WORKED MODEL 63 | KCL node with injected current

Question: A 6 A source injects into a node connected to ground through 3 Ω and 6 Ω.

1. KCL: V/3+V/6=6.
2. Multiply by 6: 2V+V=36.
3. V=12 V; currents are 4 A and 2 A.

> **RESULT Answer:** Node voltage=12 V.

WORKED MODEL 64 | Capacitive current in AC

Question: A 10 μF ideal capacitor is connected across 100 V RMS at 50 Hz.

1. IC=V/XC=ωCV.
2. IC=2π×50×10×10^{-6}×100≈0.3142 A.

> **RESULT Answer:** 0.314 A RMS, leading applied voltage by 90°.

WORKED MODEL 65 | Series-parallel stored energy

Question: Two equal 2 mH uncoupled inductors are connected in parallel and supplied with total 4 A steady current. Find total stored energy.

1. Equivalent inductance of two equal uncoupled inductors=1 mH.
2. W=½Leq Itotal^2=½×0.001×16=0.008 J.

> **RESULT Answer:** 8 mJ stored energy (assuming equal current splitting).

WORKED MODEL 66 | Battery maximum power model

Question: A 9 V cell has internal resistance 1 Ω. Choose the purely resistive external load for maximum terminal power.

1. The battery has Thevenin E=9 V and Rth=1 Ω.
2. Max power transfer when RL=Rth=1 Ω.
3. I=9/(1+1)=4.5 A; PL=I^{2}RL=20.25 W.

> **RESULT Answer:** RL=1 Ω; idealized maximum power 20.25 W. In practice battery thermal/current ratings can prohibit this.

WORKED MODEL 67 | Unbalanced source phases: caution

Question: Three phase-to-neutral currents have phasors 3∠0°, 3∠−120° and 3∠120° A. Find neutral current.

1. The three currents form a balanced set.
2. Phasor sum IA+IB+IC=0.

> **RESULT Answer:** Neutral current is zero (assuming defined return reference).

13.4 Careful negative-sign interpretation

Suppose you assumed branch current to the right and computed $I=-2$ A. The physical current is 2 A left. For a voltage reference marked + on left and − on right, $V=-10$ V means the right terminal is at a potential 10 V higher than left. A negative reactive power Q represents capacitive behavior under the passive sign convention; it is not an error.

14 High-yield theory checkpoints

14.1 Which facts should be remembered exactly?

Topic	What SSC JE can test
Units	1 W=1 J/s; 1 kWh=3.6 MJ; 1 Ω=1 V/A; 1 F=1 C/V; 1 H=1 V·s/A
KCL	Signed sum of currents at a node equals zero
KVL	Signed sum of voltage changes around an ideal lumped loop equals zero
Resistor	Energy dissipation; no ideal electromagnetic energy-storage state
Capacitor	Voltage continuity; DC steady-state open; leading sinusoidal current
Inductor	Current continuity; DC steady-state short; lagging sinusoidal current
Series resonance	Minimum |Z| and maximum I (fixed V); voltage magnification
Parallel resonance	Large |Zin| and small input current for an ideal tank at fixed V
Magnetics	Flux analogous to current; MMF analogous to voltage; reluctance analogous to R
AC complex power	S=VI*, with V,I RMS; sign of Q gives lead/lag
3-phase	P=√3 VL IL pf for balanced loads

14.2 Source and polarity reasoning

- Voltage source in parallel with another ideal voltage source is inconsistent if their imposed voltages disagree; do not solve contradictory ideal-source networks as if they were ordinary resistor circuits.
- Current source in series with another unequal ideal current source is inconsistent unless a valid alternative current path exists.
- An open-circuited ideal voltage source can still impose a terminal voltage; a short-circuited ideal current source can still impose current.
- A passive resistor always has nonnegative absorbed power for real R>0.
- When using a capacitor or inductor initial condition, pay attention to a given pre-existing voltage or current; “at switch-on” does not always mean zero.

14.3 Solve numericals without a calculator when possible

Factor numbers before decimal calculations; use Pythagorean triples for impedance triangles (3-4-5, 5-12-13, 6-8-10, 8-15-17), and remember √3≈1.732 and √2≈1.414. If values permit, keep exact expressions with π and square roots until the final step. Use 1 kΩ × 1 μF=1 ms and 1 MΩ × 1 μF=1 s for time constants.

WORKED MODEL 68 | Mental impedance triangle

Question: A series circuit has R=9 Ω and net inductive X=12 Ω; supply 150 V RMS.

1. Recognize 9-12-15 triangle: |Z|=15 Ω.
2. I=150/15=10 A.
3. PF=9/15=0.6 lagging; P=150×10×0.6.

> **RESULT Answer:** I=10 A and P=900 W.

15 All-in-one formula revision sheets

15.1 Basic quantities and DC networks

Theme	Must-know equations
Charge/current/power	$I=dq/dt$; $P=VI=I^2R=V^2/R$; $E=Pt$; 1 kWh=3.6×10^6 J
Resistive material	$R=\rho\ell/A$; $R_2=R_1(1+\alpha\Delta T)$; $G=1/R$; $\sigma=1/\rho$
Series/parallel	$R_{series}=\Sigma R$; $1/R_{parallel}=\Sigma 1/R$; 2-parallel: $R_1R_2/(R_1+R_2)$
Divider	$V_{out}=V_{in}\,R_{bottom}/(R_{top}+R_{bottom})$; $I_1=I_{total}\,R_2/(R_1+R_2)$
DC battery	$I=E/(r+R_L)$; $V_t=E-Ir$ on discharge
Thevenin/Norton	$V_{th}=V_{oc}$; $I_N=I_{sc}$; $V_{th}=I_N R_{th}$
Maximum power (DC)	$R_L=R_{th}$; $P_{L,max}=V_{th}^2/(4R_{th})$
Capacitor	$C=Q/V$; $i=C\,dv/dt$; $W=\frac{1}{2}CV^2$; $C_{parallel}=\Sigma C$; $1/C_{series}=\Sigma 1/C$
Inductor	$v=L\,di/dt$; $W=\frac{1}{2}LI^2$; $M=k\sqrt{(L_1L_2)}$

15.2 Magnetism and induction

Theme	Must-know equations
Magnetic flux	$B=\Phi/A$; $F=NI$; $\Re=\ell/(\mu A)$; $\Phi=NI/\Re$
Fields	Long wire: $B=\mu I/(2\pi r)$; solenoid: $B\approx\mu NI/\ell$
Faraday and motion	$e=-N\,d\Phi/dt$; $\lvert e\rvert=B\ell v$ for mutual perpendicular motion
Inductance	$L\approx N^2/\Re$; $M=k\sqrt{(L_1L_2)}$; Laiding=L_1+L_2+2M

15.3 Alternating current, impedance, resonance

Theme	Must-know equations
Frequency	$\omega=2\pi f$; $T=1/f$
Sinusoidal wave	$V_{rms}=V_m/\sqrt{2}$; rectified $V_{avg}=2V_m/\pi$; form factor≈ 1.11
Impedance	$Z_R=R$; $Z_L=j\omega L$; $Z_C=1/(j\omega C)$; $Z_{series}=R+j(X_L-X_C)$
Power	$S=VI^*=P+jQ$; $P=VI\cos\varphi$; $Q=VI\sin\varphi$; $\lvert S\rvert=VI$
PF improvement	$Q_c=P(\tan\varphi_1-\tan\varphi_2)$; $C_1\varphi=Q_c/(\omega V^2)$
Resonance	$\omega_0=1/\sqrt{(LC)}$; $f_0=1/(2\pi\sqrt{LC})$
Series Q and BW	$Q=\omega_0 L/R$; $BW=R/(2\pi L)$; $Q=f_0/BW$
Parallel ideal tank	$B=\omega C-1/(\omega L)$; $B=0$ at resonance

15.4 Three-phase systems

Theme	Must-know equations
Balanced star	$V_L=\sqrt{3}V_{ph}$, $I_L=I_{ph}$
Balanced delta	$V_L=V_{ph}$, $I_L=\sqrt{3}I_{ph}$
Power	$P=\sqrt{3}V_L I_L\cos\varphi$; $Q=\sqrt{3}V_L I_L\sin\varphi$; $\lvert S\rvert=\sqrt{3}V_L I_L$
Balanced star/delta equivalent	$Z_\Delta=3Z_Y$
Same three resistors moved Y→Δ	$P_\Delta=3P_Y$ at the same V_L
Unbalanced star, connected neutral	$I_N=-(I_A+I_B+I_C)$ as phasors

15.5 Time-domain optional formulas

Theme	Must-know equations
First-order template	$x(t)=x_f+(x_0-x_f)e^{(-t/\tau)}$
RC time constant	$\tau=R_{eq}C$
RL time constant	$\tau=L/R_{eq}$
RC charging	$v_C(t)=V_s(1-e^{(-t/RC)})$
RL current rise	$i_L(t)=(V_s/R)(1-e^{(-tR/L)})$

16 Topic-wise practice set: objective models

Attempt this set as a timed test after completing the theory. These are newly written practice questions, **not claimed SSC previous-year questions**. Answer key with short reasoning follows. All quantities use ideal components unless otherwise stated.

PRACTICE QUESTIONS

DC fundamentals - Questions

1. A constant current of 2 A flows for 5 minutes. Charge moved equals:

A. 120 C B. 300 C C. 600 C D. 1200 C

2. A 100 W lamp is operated 10 hours. Energy in kWh:

A. 0.1 B. 1 C. 10 D. 100

3. One unit of electrical energy on a typical electricity bill is:

A. 1 J B. 1 kJ C. 1 kWh D. 1 W

4. If a conductor is stretched to twice its original length at constant volume, R becomes:

A. 2R B. 4R C. R/2 D. R/4

5. A 40 Ω wire is divided into four equal lengths and all four parts are connected in parallel. Equivalent R:

A. 10 Ω B. 2.5 Ω C. 40 Ω D. 160 Ω

6. At a node, 7 A enters and two branch currents leave, one 3 A. The other is:

A. 4 A entering B. 4 A leaving C. 10 A entering D. 10 A leaving

7. 3 Ω and 6 Ω are in parallel; the equivalent resistance is:

A. 9 Ω B. 3 Ω C. 2 Ω D. 18 Ω

8. For three identical 9 Ω resistors in parallel, Req is:

A. 27 Ω B. 9 Ω C. 6 Ω D. 3 Ω

9. Which is always zero across an ideal short circuit?

A. Current B. Power delivered by network C. Voltage D. Resistance of all nearby branches

10. A resistor carries 3 A and its terminal voltage is 12 V under passive sign convention. It absorbs:

A. 4 W B. 15 W C. 36 W D. 144 W

11. Two series resistors 2 Ω and 8 Ω across 50 V: voltage across 8 Ω is:

A. 10 V B. 20 V C. 40 V D. 48 V

12. What is the ideal internal resistance of a voltage source?

A. 0 Ω B. ∞ Ω C. 1 Ω D. Load resistance

13. A balanced Wheatstone bridge carries what current through its bridging arm?

A. Maximum B. Zero C. Half source current D. Infinite

14. An initially open 5 Ω branch in a network has current:

A. 0 A B. 1 A C. 5 A D. Cannot be zero

Capacitors & inductors - Questions

15. Two 4 µF capacitors connected in parallel give:

A. 2 µF B. 4 µF C. 8 µF D. 16 µF

16. Two 4 µF capacitors in series give:

A. 2 µF B. 4 µF C. 8 µF D. 16 µF

17. An ideal capacitor connected to steady DC ultimately behaves as:

A. Short B. Open C. Voltage source of zero volts D. Current source

18. An ideal inductor at DC steady state is:

A. Open B. Short C. Equal to 1 H resistor D. A capacitor

19. Energy in 2 F capacitor at 3 V is:

A. 3 J B. 6 J C. 9 J D. 18 J

20. An ideal 2 H inductor carries 4 A. Stored energy:

A. 4 J B. 8 J C. 16 J D. 32 J

21. An isolated charged capacitor gets dielectric $\varepsilon r=3$. Its voltage becomes:

A. 3V B. V/3 C. V D. 9V

22. A capacitor connected to a constant voltage source gets $\varepsilon r=2$ inserted fully. Its stored energy:

A. Halves B. Doubles C. Unchanged D. Becomes zero

23. Series uncoupled inductances 2 H and 3 H equivalent:

A. 1.2 H B. 5 H C. 6 H D. 0.5 H

24. Coils 4 H, 9 H have coupling k=1. Mutual inductance:

A. 6 H B. 13 H C. 36 H D. 5 H

25. An ideal current source has internal resistance:

A. Zero B. Infinite C. Exactly one ohm D. Same as load

26. A discharging battery has E=12 V, r=1 Ω, I=2 A. Terminal voltage:

A. 14 V B. 12 V C. 10 V D. 6 V

Network analysis - Questions

27. Nodal analysis is fundamentally based on:

A. KCL B. KVL alone C. Faraday law D. Coulomb friction

28. Mesh analysis primarily applies:

A. KVL B. KCL alone C. Stefan law D. Gauss law exclusively

29. To deactivate an independent ideal current source, replace it by:

A. Short B. Open C. Inductor D. Dependent source

30. To deactivate an independent ideal voltage source, replace it by:

A. Open B. Short C. Capacitor D. Resistor of 10 Ω

31. During superposition, the following remain active:

A. Only resistors B. Dependent sources C. No sources D. Every independent source

32. Superposition may directly add:

A. Individual-source resistor powers B. Individual-source signed branch voltages C. Squares of currents D. Efficiencies

33. A Thevenin circuit consists of:

A. Current source parallel R B. Voltage source series R C. Inductor only D. Two current sources

34. A Norton circuit consists of:

A. Voltage source series R B. Current source parallel R C. Current source series R D. Open circuit only

35. A 20 V Thevenin source with Rth=5 Ω transfers max power to RL equal to:

A. 2.5 Ω B. 5 Ω C. 10 Ω D. 20 Ω

36. Max resistor load power for Vth=20 V, Rth=5 Ω is:

A. 10 W B. 20 W C. 40 W D. 80 W

37. An AC Thevenin impedance is 6+j8 Ω. Optimum unconstrained complex load is:

A. 6+j8 B. 6−j8 C. 8−j6 D. 10+j0

38. Ideal current source 4 A parallel 2 Ω converts to a source voltage of:

A. 2 V B. 4 V C. 8 V D. 16 V

Magnetic circuits - Questions

39. Unit of magnetic flux is:

A. Tesla B. Weber C. Henry D. Farad

40. Unit of magnetic flux density is:

A. Weber B. Volt C. Tesla D. Ohm

41. Magnetomotive force of N turns carrying I is:

A. NI B. N/I C. I/N D. I^2R

42. Reluctance for uniform magnetic path is:

A. $\mu A/\ell$ B. $\ell/(\mu A)$ C. NI D. B/H

43. The addition of an air gap usually:

A. Decreases total reluctance B. Increases total reluctance C. Makes reluctance zero D. Removes hysteresis completely

44. Faraday induced EMF magnitude depends on:

A. Flux alone B. Rate of change of flux linkage C. Resistor color D. Mechanical torque only

45. Lenz law says induced EMF:

A. Aids flux change B. Opposes the change creating it C. Is always zero D. Always increases current

46. Narrow hysteresis loop is preferred for:

A. Permanent magnet B. Transformer core C. Permanent magnetization storage D. Compass needle

47. Magnetic field near a long straight conductor varies with distance r as:

A. r B. r^2 C. 1/r D. Independent of r

48. At constant flux and fixed area, flux density is:

A. $B=\Phi/A$ B. $B=\Phi A$ C. $B=A/\Phi$ D. B=0

AC circuits - Questions

49. For a sine wave Vrms is:

A. Vm B. $Vm/\sqrt{2}$ C. $2Vm/\pi$ D. Vm/2

50. Average sinusoidal voltage over a complete cycle:

A. $Vm/\sqrt{2}$ B. $2Vm/\pi$ C. 0 D. Vm

51. The full-wave rectified mean of a sine wave is:

A. 0 B. $Vm/\sqrt{2}$ C. $2Vm/\pi$ D. Vm

52. Period of a 25 Hz AC quantity:

A. 4 ms B. 20 ms C. 40 ms D. 50 ms

53. If frequency doubles, ideal inductive reactance:

A. Halves B. Doubles C. Unchanged D. Becomes zero

54. If frequency doubles, ideal capacitive reactance:

A. Halves B. Doubles C. Unchanged D. Becomes infinite

55. For ideal resistor current:

A. Leads V by 90° B. Lags V by 90° C. In phase with V D. Opposes V by 180°

56. For ideal inductor current:

A. Leads 90° B. Lags 90° C. In phase D. Has no phase

57. For ideal capacitor current:

A. Leads 90° B. Lags 90° C. In phase D. Leads 180°

58. A series $Z=3+j4\ \Omega$ has magnitude:

A. 1 Ω B. 5 Ω C. 7 Ω D. 12 Ω

59. A series load has $R=6\ \Omega$ and net capacitive reactance $-8\ \Omega$. Its PF:

A. 0.6 leading B. 0.6 lagging C. 0.8 leading D. 0.8 lagging

60. For sinusoidal RMS phasors, complex power is:

A. VI B. VI* C. V/I D. I/V

61. A 5 kVA load at 0.8 PF has real power:

A. 2 kW B. 3 kW C. 4 kW D. 5 kW

62. Negative Q under passive sign convention means:

A. Inductive B. Capacitive C. Always no load D. Negative resistance

63. Parallel R and ideal C across a sinusoidal source produce total supply current that:

A. Leads source voltage B. Lags source voltage C. Is exactly zero D. Is always 180° out of phase

64. A load has 1000 W and 500 var (positive). Apparent power is:

A. 500 VA B. 1000 VA C. ≈1118 VA D. 1500 W exactly

Resonance - Questions

65. At series resonance, net reactance is:

A. Largest positive B. Zero C. Largest negative D. Infinite

66. A series RLC at resonance has PF:

A. 0 leading B. 0 lagging C. 1 D. Cannot be defined

67. At ideal series resonance, input impedance:

A. R B. Zero for R>0 C. Infinite D. ωL

68. At resonance a series RLC connected to fixed voltage has:

A. Minimum current B. Maximum current C. Zero current always D. No capacitor voltage

69. If L becomes 4L and C unchanged, ideal f0 becomes:

A. 4f0 B. 2f0 C. f0/2 D. f0/4

70. For series RLC, quality factor is:

A. $R/(\omega 0L)$ B. $\omega 0L/R$ C. LC D. L/R^2

71. Series RLC bandwidth in hertz is:

A. $R/(2\pi L)$ B. $1/(2\pi RC)$ C. R/L D. $2\pi L/R$

72. At a series half-power point current is:

A. Imax B. Imax/2 C. Imax/√2 D. √2 Imax

73. Ideal parallel LC impedance at resonance approaches:

A. 0 B. R of ideal wire C. ∞ D. √(L/C) always

74. At fixed V, series RLC below resonance behaves:

A. Inductively B. Capacitively C. Resistively only D. As an ideal short

Three-phase - Questions

75. Balanced phase displacement in a 3-phase system:

A. 60° B. 90° C. 120° D. 180°

76. In balanced star, line current equals:

A. Phase current B. √3 phase current C. Phase current/√3 D. Zero

77. In balanced delta, phase voltage equals:

A. VL/√3 B. VL C. √3 VL D. 0

78. In balanced star, line voltage is:

A. Vph B. Vph/√3 C. √3 Vph D. 3Vph

79. In balanced delta, line current magnitude equals:

A. Iph B. Iph/√3 C. √3 Iph D. 3Iph

80. Total balanced 3-phase active power is:

A. VL IL cosφ B. 3VL IL cosφ C. √3 VL IL cosφ D. VL IL/√3

81. Balanced 3-phase, VL=400 V, IL=10 A, unity PF: P approximately:

A. 2.31 kW B. 4 kW C. 6.93 kW D. 12 kW

82. Balanced star load has 10 Ω per phase. Equivalent balanced delta impedance per phase:

A. 10/3 Ω B. 10 Ω C. 20 Ω D. 30 Ω

83. Same identical 3 resistors shifted from star to delta at fixed line voltage: total power:

A. Unchanged B. Multiplies by 3 C. Divides by 3 D. Becomes zero

84. A balanced Y load has neutral current:

A. Equal to line current B. √3×line current C. Zero D. Infinite

85. A delta phase branch carries 5 A RMS; line current magnitude is:

A. 5/√3 A B. 5 A C. 5√3 A D. 15 A

86. When asked to use √3VLILpf, the load should normally be:

A. Any unbalanced circuit B. Balanced 3-phase C. Single-phase only D. A DC load

First-order response - Questions

87. RC time constant:

A. R/C B. RC C. L/R D. 1/RC

88. RL time constant:

A. RL B. R/L C. L/R D. 1/RL

89. After 1τ, charging capacitor voltage from zero reaches approximately:

A. 36.8% B. 50% C. 63.2% D. 100%

90. After 1τ, a decaying RC voltage remains at:

A. 13.5% B. 36.8% C. 63.2% D. 95%

91. Inductor current immediately after finite-voltage switching:

A. Always 0 B. Equals just-before value C. Instantly becomes infinity D. Always final value

92. Capacitor voltage immediately after finite-current switching:

A. Always 0 B. Always source voltage C. Equals just-before value D. Always infinity

93. 10 kΩ and 100 µF give τ:

A. 1 ms B. 10 ms C. 100 ms D. 1 s

94. An RC charge reaches half of its final change after:

A. τ B. τln2 C. 2τ D. 5τ

17 Practice answer key with short explanations

Use the explanation to detect a wrong method, not merely to confirm the selected letter. The section tag lets you return to the associated theory chapter.

Q	Ans	Reason
1	C	Q=It=2×300=600 C.
2	B	0.1 kW×10 h=1 kWh.
3	C	Billing unit is one kilowatt-hour.
4	B	Area halves, hence R scales as ℓ^2.
5	B	Req=R/n²=40/16=2.5 Ω.
6	B	KCL: 7=3+4.
7	C	Req=3×6/(3+6)=2 Ω.
8	D	9/3=3 Ω.
9	C	Ideal short has zero terminal voltage.
10	C	P=VI=12×3=36 W.
11	C	Voltage division 50×8/10=40 V.
12	A	Ideal voltage source has zero internal series R.

Q	Ans	Reason
13	B	Midpoints have equal potentials.
14	A	Open branch current is zero.
15	C	Parallel capacitances add.
16	A	Two equal series capacitors halve.
17	B	i=C dv/dt; dv/dt=0 at DC steady state.
18	B	v=L di/dt=0 at constant current.
19	C	½CV²=½×2×9=9 J.
20	C	½×2×16=16 J.
21	B	Isolated means Q fixed; C triples.
22	B	At fixed V, W=½CV² and C doubles.
23	B	Inductances add in series if not coupled.
24	A	M=k√(4×9)=6 H.

Q	Ans	Reason
25	B	Ideal current source has infinite parallel output resistance.
26	C	$E-Ir=12-2=10$ V.
27	A	Node equations conserve current.
28	A	Loop sums satisfy KVL.
29	B	Set current to zero: open.
30	B	Set voltage to zero: short.
31	B	Keep dependent sources active.
32	B	Linear variables superpose; power does not.
33	B	Thevenin source in series with Rth.
34	B	Norton source in parallel with RN.
35	B	Resistive matching RL=Rth.
36	B	$P_{max}=400/(4\times5)=20$ W.

Q	Ans	Reason
37	B	Conjugate match.
38	C	$V_{th}=IR=4\times2=8$ V.
39	B	Flux is measured in Wb.
40	C	B is measured in tesla = Wb/m^2.
41	A	MMF=ampere-turns NI.
42	B	Magnetic Ohm law: $\Re=\ell/(\mu A)$.
43	B	Air has much smaller permeability than iron.
44	B	$\lvert e\rvert=\lvert d(N\Phi)/dt\rvert$.
45	B	Opposes change in flux linkage.
46	B	Soft magnetic core reduces hysteresis loss.
47	C	$B=\mu I/(2\pi r)$.
48	A	Definition $B=\Phi/A$ for uniform normal flux.

Q	Ans	Reason
49	B	$V_{rms}=peak/\sqrt{2}$.

Q	Ans	Reason
50	C	Symmetric positive and negative halves cancel.
51	C	Rectified mean=2Vm/π.
52	C	1/25=0.04 s=40 ms.
53	B	XL=2πfL.
54	A	XC=1/(2πfC).
55	C	Resistive current and voltage are in phase.
56	B	I=V/(jXL) gives −90°.
57	A	I=jωCV has +90° angle.
58	B	Magnitude sqrt(3²+4²)=5 Ω.
59	A	|Z|=10, PF=6/10=0.6 leading.
60	B	S=V I* with conjugate of I.

Q	Ans	Reason
61	C	P=5×0.8=4 kW.
62	B	Capacitive reactive power is negative.
63	A	Capacitor adds positive imaginary current.
64	C	sqrt(1000²+500²)=1118 VA.
65	B	XL−XC=0.
66	C	Purely resistive input impedance.
67	A	Net reactive parts cancel, leaving R.
68	B	Impedance minimum R.
69	C	f0∝1/√L.
70	B	Series Q=ω0L/R.
71	A	Δf=R/(2πL).
72	C	Power∝I².

Q	Ans	Reason
73	C	Inductor and capacitor admittances cancel.
74	B	Below f0: XC>XL.

Q	Ans	Reason
75	C	Phase spacing 120°.
76	A	IL=Iph in Y.
77	B	Vph=VL in Δ.
78	C	VL=√3Vph.
79	C	IL=√3Iph.
80	C	P=√3VLILpf.
81	C	√3×400×10≈6.928 kW.
82	D	ZΔ=3ZY.
83	B	At fixed VL, each Δ resistor gets √3 times former Y voltage.
84	C	Balanced current phasors sum to zero.

Q	Ans	Reason
85	C	IL=√3Iph.
86	B	Expression presumes balanced 3-phase line quantities.
87	B	τ=RC.
88	C	τ=L/R.
89	C	$1-e^{-1}\approx0.632$.
90	B	$e^{-1}\approx0.368$.
91	B	iL(0+)=iL(0−).
92	C	vC(0+)=vC(0−).
93	D	$10{,}000\times100\times10^{-6}=1$ s.
94	B	Half-step occurs at τln2.

18 Final checklist, mock strategy and common errors

18.1 Concept checklist: tick only when you can solve independently

- ☐ I can calculate charge/energy, physical resistance changes and loaded dividers without mixing units.
- ☐ I can identify series/parallel from node connections, not circuit drawing appearance.
- ☐ I can convert delta to star and star to delta with correct terminal labels.
- ☐ I can calculate capacitor series voltage division and energy under constant Q versus constant V.
- ☐ I can use dot convention, coupled inductors and practical-battery terminal voltage.
- ☐ I can write node equations, use a supernode and solve two-mesh equations.
- ☐ I know the exact source-killing rule and the treatment of dependent sources.
- ☐ I can obtain Vth, Rth, IN, RN and load power at a specified port.
- ☐ I can distinguish adding voltages/currents from the invalid addition of powers in superposition.
- ☐ I can solve magnetic circuits using total MMF drops including air gaps.
- ☐ I know full-cycle sine average versus rectified mean; and peak-to-RMS conversion.
- ☐ I can convert R, L, C to phasor impedances, analyze both series and parallel AC circuits.
- ☐ I can correctly assign lagging/leading sign and calculate complex power.
- ☐ I can compute resonance frequency, Q, bandwidth and individual reactive voltages.
- ☐ I can distinguish series resonance from a parallel tank.
- ☐ I can identify star/delta phase and line voltage/current relations before finding 3-phase power.
- ☐ I can correct lagging power factor with a shunt capacitor (single phase and 3-phase connection-aware).

18.2 Most avoidable numerical mistakes

Mistake	Correction
Adding AC voltage magnitudes directly	Add phasors (rectangular/complex form).
Using crest values in power formulas	Use RMS V and I for ordinary AC power.
Using 1/2 CV instead of 1/2 CV^2	Remember capacitor energy is quadratic in voltage.
Using current divider same as voltage divider	Current is inversely related to parallel branch R.
Neglecting loading of a potential divider	Replace lower R by R||RL first.
Assuming source killed means source resistance removed	Ideal V source becomes short, ideal I source becomes open; keep remaining elements.
Killing a dependent source for Rth	Leave it active and apply a test source.
Applying RL=Rth to every AC load	Use complex conjugate match when load is a free complex impedance.
Assuming induced EMF always positive	Lenz-law sign depends on chosen polarity.
Confusing √3 connection factors	Y: $V_L=\sqrt{3}V_{ph}$. Δ: $I_L=\sqrt{3}I_{ph}$.
Using source angle for PF	PF uses angle between load voltage and load current.
Assuming capacitor initially open or inductor initially short	Those are DC final-state models, not general 0+ models.

18.3 Suggested 4-session revision cycle

Session	Study + practice focus
1	Chapters 1-4: resistance, capacitors, inductors, basic sources. Redo models 1-13 and corresponding questions.
2	Chapters 5-7: node/mesh, theorems, magnetic circuits. Solve every worked network from scratch.
3	Chapters 8-11: RMS, series/parallel AC, power, resonance, optional transients. Review phasor signs.
4	Chapters 12-18: 3-phase, formula sheets, complete MCQs, corrections and final concept checklist.

EXAM USAGE For Paper I, prioritize rapid recognition and elimination; for Paper II, work through full equations and verify physical meaning. No fixed chapter-wise marks distribution is guaranteed by SSC.

18.4 Sources and scope declaration

Primary official source: Staff Selection Commission, Notice - Junior Engineer Examination, 2026, published 02 September 2026; paragraph 14.3.2 “Part-B (Electrical Engineering)”, printed page 26 of 97: https://ssc.gov.in/api/attachment/uploads/masterData/NoticeBoards/Notice_of_adv_je_2026.pdf

This book is independently composed for self-study and expands the official headings into standard undergraduate/diploma-level models. Sections explicitly marked as extensions are supplementary, rather than claims that SSC expressly names them in the 2026 syllabus. Model questions are created for practice; no claim is made that they are verbatim previous-year SSC questions.

END OF STUDY GUIDE